HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

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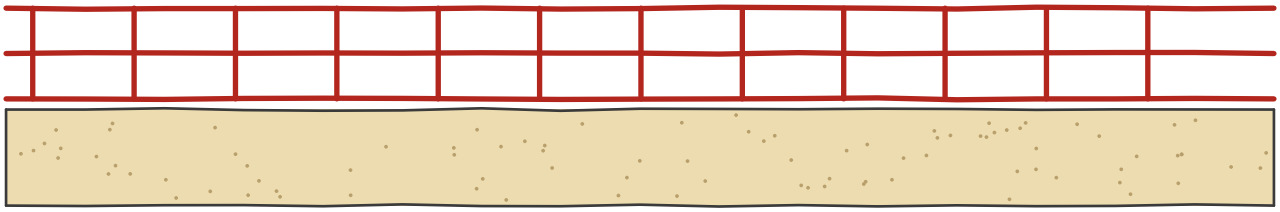
UNIT – I

Reinforced Earth

Syllabus topics covered in these notes:

- Reinforced Earth: concept, effects of reinforcement on soils
- Equal (enhanced) confining pressure and pseudo-cohesion concepts
- Materials; friction coefficient: definition, laboratory determination, factors affecting it
- Applications of reinforced earth

soil + reinforcement = strong composite



1. Reinforced Earth – Concept

Reinforced earth (Terre Armée) is a **composite material** made of compacted frictional soil + tension-resisting reinforcement (steel strips, geogrids, geotextiles) placed in layers. It was invented by the French architect-engineer **Henri Vidal** in the 1960s; the first structure (a wall) was built at Pragnères, France, in 1965. Today it is also called **MSE** (Mechanically Stabilised Earth).

- **1963–65:** Vidal patented the system; first wall at Pragnères (Pyrenees).
- **1972:** first wall in the USA (California, SR-39).
- **1970s–80s:** geotextile and then geogrid reinforcement introduced; now widely used worldwide, including India (flyover approaches, bridge abutments, hill roads).

Soil is strong in compression & shear but very weak in tension. Reinforcement is strong in tension. When combined, the soil mass behaves like a block of RCC: soil takes compression, reinforcement takes tension.

Remember

Analogy: RCC = concrete + steel bars

Reinforced earth = soil + strips / grids / sheets

Basic mechanism

- Soil mass tries to deform laterally when loaded (vertical load, self-weight, surcharge).
- Reinforcement is placed horizontally, in the direction of tensile strain, so soil tries to move relative to the strip.
- Friction (and passive bearing for grids) between soil and reinforcement develops **tension T** in the reinforcement.
- Lateral movement is restrained, so soil acts as if it were under higher confinement; the mass becomes **coherent**, stiff and strong.

Components of a reinforced earth structure

1. **Backfill** – compacted granular, free-draining frictional soil.

2. **Reinforcement** – metallic or polymeric tensile elements in horizontal layers.
3. **Facing** – precast concrete panels, modular blocks, wrap-around geosynthetic, gabion, welded wire. It only prevents local sloughing; it carries little load.
4. **Foundation / leveling pad** and **drainage** system.

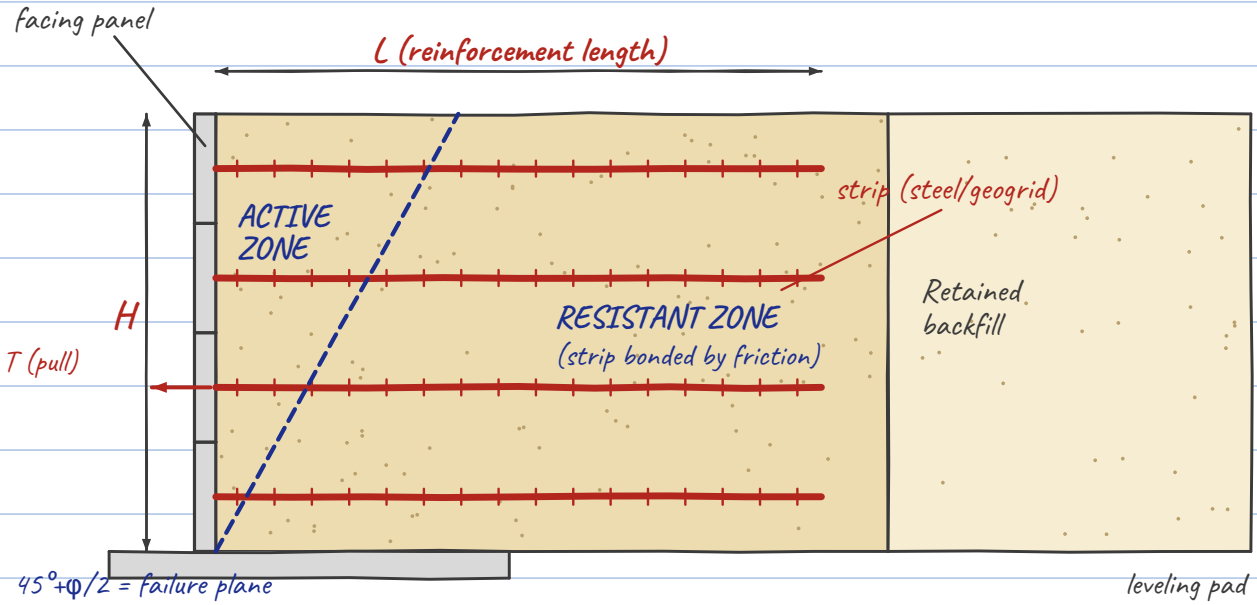


Fig 1.1 Reinforced earth wall – strips, facing and failure plane

Reinforced earth vs RCC

Property	RCC	Reinforced earth
Matrix	Concrete (cement bound)	Soil (friction bound)
Reinforcement	Steel bars, bond by adhesion	Strips / grids, bond by friction
Behaviour	Rigid, brittle	Flexible, ductile
Settlement	Cracks easily	Tolerates differential settlement

2. Effects of Reinforcement on Soils

- Increases **shear strength** and **bearing capacity** of the soil mass.
- Increases **stiffness** (modulus) and reduces **settlement**.
- Restrains **lateral strain** (lateral spreading) and horizontal deformations.
- Improves **ductility**: failure is gradual, not sudden; post-peak strength loss is reduced.

- Results in **anisotropic** strength: strong along reinforcement direction, weaker across it.
- Allows **steeper slopes** and vertical walls with granular soil.

Two classical concepts explain the strength gain (Yang 1972; Schlosser and Long 1972; Hausmann 1976):

(a) Equal (enhanced) confining pressure concept

Reinforcement restrains the lateral expansion of soil. This restraint is equivalent to an **additional confining pressure** $\Delta\sigma_3$ acting on the soil, without any change in its friction angle φ.

$$\Delta\sigma_3 = T / (S_v \cdot S_h) \text{ (strips of spacing } S_v \times S_h)$$
$$\Delta\sigma_3 = T / S_v \text{ (continuous sheets, } T \text{ per metre width)}$$

where T = tension mobilised in the reinforcement at failure (per strip / per metre).

At failure of the soil (Rankine/Mohr–Coulomb):

$$\sigma_1 = (\sigma_3 + \Delta\sigma_3) K_p, \quad K_p = \tan^2(45° + \varphi/2)$$
$$\Delta\sigma_1 = \Delta\sigma_3 \cdot K_p$$

So the Mohr circle becomes **bigger** for the same σ_3, but the failure envelope of the soil (same φ) is unchanged. Triaxial tests (Yang 1972) show: strength increases with number of layers, closer spacing and higher modulus of reinforcement.

(b) Pseudo-cohesion (apparent cohesion) concept

Here the reinforced soil is treated as a **new anisotropic material** that has the same φ but an **apparent cohesion** c_R – even for a cohesionless sand. The failure envelope shifts upward by c_R.

Derivation: for the reinforced soil $\sigma_1 = \sigma_3 K_p + 2c_R\sqrt{K_p}$. Equating with $\sigma_1 = \sigma_3 K_p + \Delta\sigma_3 K_p$ gives

$$c_R = (\Delta\sigma_3 / 2) \cdot \sqrt{K_p} = [T / (2 S_v S_h)] \cdot \tan(45° + \varphi/2)$$

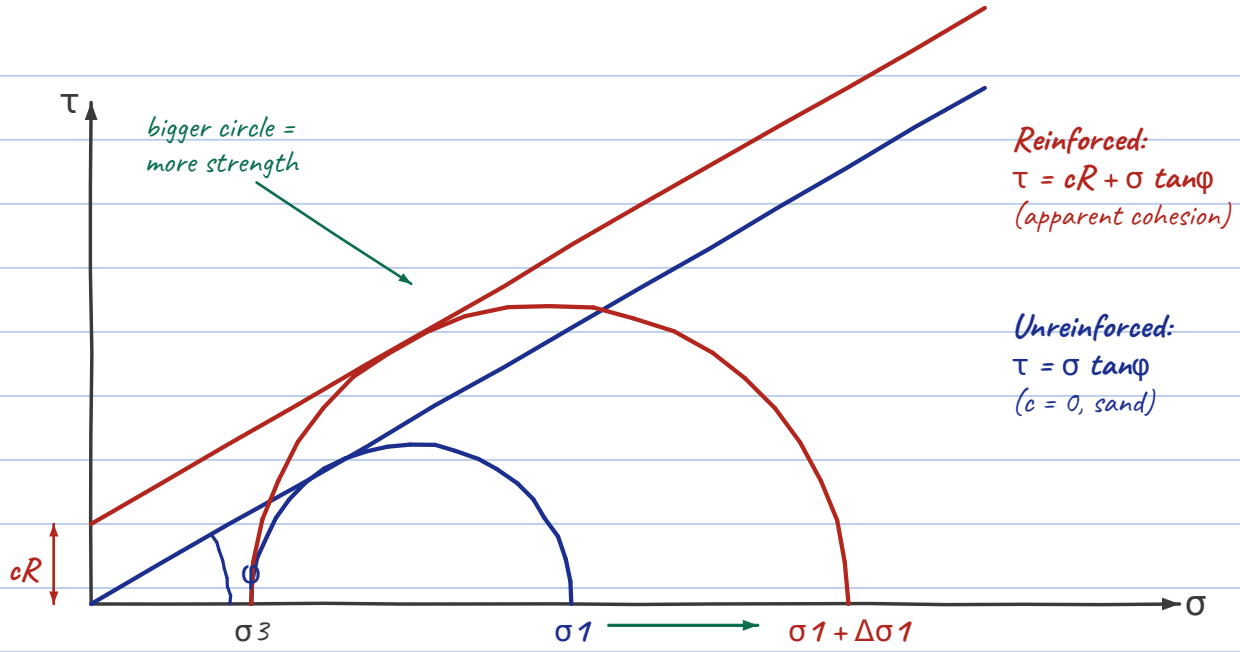


Fig 1.2 Mohr circles – effect of reinforcement

Example 1

Worked example: sand $\varphi = 30^\circ$, reinforcement tension $T = 15$ kN per strip, $S_v = 0.5$ m, $S_h = 0.75$ m.

$\Delta\sigma_3 = 15 / (0.5 \times 0.75) = 40$ kPa; $K_p = 3$; $\sqrt{K_p} = 1.732$

$c_R = (40 / 2) \times 1.732 = 34.6$ kPa; $\Delta\sigma_1 = 40 \times 3 = 120$ kPa

Comparison of the two concepts

Point	Enhanced confining pressure	Pseudo-cohesion
Idea	Extra σ3 on same soil	New material with cohesion c_R
φ of soil	Unchanged	Unchanged
Envelope	Same line; bigger circle	Line shifted up by c_R
Best for	Closely spaced sheets, triaxial	Strips, design of reinforced beds

3. Materials for Reinforced Earth

3.1 Backfill (reinforced fill)

- Must be **granular, free draining**, with good friction angle $\varphi \geq 30^\circ$–34°.
- Fines passing 75 μm sieve ≤ **15 %**; plasticity index PI ≤ **6**; maximum particle size ≤ 75 mm (to avoid damage to reinforcement).

- Compaction to ≥ **95 %** of maximum dry density in 150–200 mm lifts; use light equipment near the facing.
- For steel strips, backfill should be **non-corrosive**: pH 5–10, resistivity ≥ 3000 Ω·cm, chlorides ≤ 100 ppm, sulphates ≤ 200 ppm, organic content ≤ 1 %.
- Clays, silts, organic soils and frozen soils should be avoided: low friction, poor drainage, pore pressure build-up.

3.2 Reinforcement

- **Metallic strips** – galvanised mild steel ribbed strips, e.g. 50 mm × 4 mm, zinc coating ≈ 70–85 µm. Inextensible, high modulus, high pull-out strength.
- **Steel grids / welded wire mesh / bar mats** – friction + passive resistance on cross bars.
- **Geosynthetics** – geotextiles (woven), geogrids (HDPE, PP, PET), geostrips (polyester yarn in PE sheath). Extensible, non-corrosive, light and easy to handle.
- Ideal properties: high tensile strength, low creep, good soil friction, durable, ductile, easily jointed and cheap.

Note

Corrosion design (galvanised steel, FHWA):

Zinc loss ≈ 15 µm/year for first 2 years, then ≈ 4 µm/year; carbon steel afterwards ≈ 12 µm/year.

Sacrificial thickness is added to the design life (75–100 years for permanent walls).

Extensible vs inextensible reinforcement

Property	Inextensible (steel)	Extensible (geosynthetic)
Strain at failure	< 1–2 %	10 % to 80 %
Modulus	Very high	Low to moderate
Failure surface	Bilinear (coherent gravity)	Rankine plane
Earth pressure coeff.	K0 → Ka with depth (1.7Ka→1.2Ka)	Ka (constant)
Creep	Negligible	Significant for PP / PE
Durability issue	Corrosion	UV, chemical, creep, damage

Failure modes (internal) of reinforced soil

1. **Tension (rupture) failure** – reinforcement breaks; occurs when bond is strong but strength is small. Cure: stronger / closer layers.
2. **Pull-out (slip) failure** – reinforcement slips out of resistant zone; occurs when length L_e is short or friction is low. Cure: longer / rougher reinforcement.
3. **Connection failure** – at facing, due to corrosion or poor detailing.
4. **Excessive deformation** – due to extensible reinforcement, poor compaction.

Design rule: **allowable tension** $\geq T_{max}$ **and pull-out resistance** $\geq T_{max}$ at every layer, with a factor of safety (usually 1.5).

3.3 Facing

- **Precast concrete panels** (cruciform / rectangular, ≈ 1.5 m × 1.5 m, 140–180 mm thick, M30 or higher) – most common for permanent walls.
- **Modular concrete blocks** with geogrid; **gabion** facing; **welded wire** facing with vegetation; **geotextile wrap-around** (temporary / green slopes).
- Facing gives aesthetics and prevents raveling; connection to reinforcement must be strong and corrosion-proof.

4. Soil–Reinforcement Friction Coefficient

4.1 Definition

Reinforced earth works only if the soil can **grip** the reinforcement. This gripping ability is measured by the **coefficient of friction** (μ).

$$\mu = \tau / \sigma_n = \tan\delta$$

τ = interface shear (frictional) stress, σ_n = normal stress on the reinforcement, δ = **angle of interface friction** between soil and reinforcement. The **interaction (efficiency) coefficient** compares it with the soil itself:

$$C_i = \tan\delta / \tan\varphi \ (\leq 1 \text{ for smooth}, \approx 1 \text{ or more for grids})$$

In pull-out analysis the **apparent friction coefficient** μ^* (or F^*) is used, which also includes dilatancy and passive resistance:

$$P_r = 2 \cdot b \cdot L_e \cdot \mu^* \cdot \sigma_v$$

P_r = pull-out resistance, b = strip width, L_e = embedded length in the resistant zone, σ_v = vertical effective stress at the level of the strip, factor 2 = top and bottom faces.

Important

Variation of μ^* with depth (ribbed steel strip, Schlosser):

at top $z = 0$: $\mu^* = 1.2 + \log C_u$ (≤ 2.0) → decreases linearly → at $z = 6$ m and below: $\mu^* = \tan\varphi$.

Default for geosynthetics: $F^* \approx 2/3 \tan\varphi$ (sheets / geogrids).

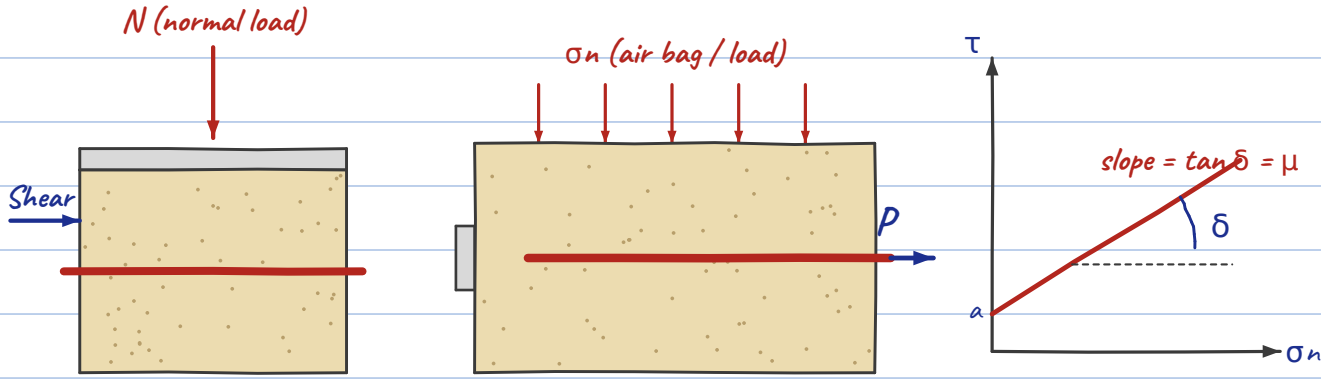
4.2 Laboratory determination

(i) Direct shear test (modified, large shear box – ASTM D5321)

- Lower half of box filled with soil (or rigid block), reinforcement fixed at the interface between the halves, upper half filled with soil compacted to field density.
- Normal load σ_n applied (3–4 values), shear displacement at about 1 mm/min.
- Peak τ plotted against σ_n; **slope = tan δ**, intercept = adhesion a.

(ii) Pull-out test (ASTM D6706) – closest to field behaviour

- Large box with reinforcement embedded in the centre of the soil; a **sleeve** at the front wall removes front-wall effects.
- Normal pressure by rubber air-bag / hydraulic jack; reinforcement pulled out at a constant rate (≈ 1 mm/min).
- Pull-out force P vs displacement recorded; peak P used: $\mu^* = P / (2\,b\,L\,\sigma_n)$.



(a) Direct shear — reinforcement at interface
(b) Pull-out test — strip length L embedded
(c) τ vs σn plot — a = adhesion intercept

Fig 1.3 Laboratory tests for friction coefficient

(iii) Inclined plane (tilt table) test – soil block placed on reinforcement fixed to a plank; plank tilted slowly; at slip angle β, $\delta = \beta$. Used for geosynthetic–soil and geosynthetic–geosynthetic interfaces; simple but low stress levels.

Examples

Example 2 (direct shear): at σ_n = 100 kPa, τ = 62 kPa → μ = 0.62 → $\delta = \tan^{-1}(0.62) = 31.8°$.

Example 3 (pull-out): P = 12 kN, b = 50 mm, L = 1.0 m, σ_n = 100 kPa → $\mu^* = 12 / (2 \times 0.05 \times 1.0 \times 100) = 1.2$

Why is μ^* larger near the top?

A dense granular soil **dilates** when sheared at the interface. The surrounding soil restrains this volume increase, so the normal stress on the strip rises above σ_v. At shallow depth confinement is small and dilatancy is large (μ^* up to 1.5–2); at greater depth the effect vanishes and $\mu^* \to \tan\varphi$.

Interface (typical, indicative)	δ or C_i
Sand – smooth steel strip	$\delta \approx$ ½ to ⅔ φ
Sand – ribbed steel strip	$\mu^* \approx$ 1.2 – 2 (top), tan φ (deep)
Sand – woven geotextile	$C_i \approx$ 0.7 – 0.9
Sand – nonwoven geotextile	$C_i \approx$ 0.8 – 1.0
Sand – geogrid	$C_i \approx$ 0.8 – 1.1

4.3 Factors affecting the friction coefficient

1. **Surface roughness / texture** of reinforcement – ribbed strips and grids give higher μ than smooth strips.
2. **Soil type and gradation** – angular, well-graded sand/gravel gives high φ and μ; fines reduce it.
3. **Particle size vs aperture** (grids) – best interlock when D_{50} is suitable relative to aperture size.
4. **Normal stress** – μ is high at low σ_n (dilatancy) and decreases with depth, approaching tan φ.
5. **Density / degree of compaction** – dense soil dilates more → higher pull-out resistance.
6. **Moisture content / saturation** – wet or saturated fines reduce friction.
7. **Reinforcement stiffness and extensibility** – extensible sheets show progressive mobilisation, so only part of length is active.
8. **Embedded length and width** – scale effects; short strips mobilise more peak μ.
9. **Rate of displacement, box size and boundary conditions** (test factors) affect the measured value.

5. Applications of Reinforced Earth

- **Retaining walls** – highway/railway, flyover and bridge approaches, ports, industrial yards (heights 3–30 m).
- **Bridge abutments** and wing walls (integral and separate).

- **Embankments on soft ground** – basal reinforcement; steep-sided reinforced embankments.
- **Reinforced soil slopes** (RSS) up to 70° – landslide repair, widening of roads in hilly areas.
- **Foundations** – reinforced soil beds under footings, rafts and storage tanks.
- **Roads, railways and airfields** – reinforced base/sub-base on weak subgrade.
- **Dams, levees, sea walls, river training** and coastal protection.
- **Others** – tunnel portals, noise barriers, mine waste dumps, military bunkers, bunds and ramps.

Advantages

- Economical (about 25–50 % cheaper than RCC walls at medium–large heights) and **fast** to build.
- Flexible: tolerates large differential settlement and has good **seismic** performance.
- No formwork, small foundations, uses local granular fill, easy to inspect and repair.
- Aesthetic (various facings) and environment friendly.

Limitations

- Needs good granular backfill and adequate space behind the face ($L \geq 0.7H$).
- Durability of metal (corrosion) / polymers (creep, UV) must be checked.
- Design is sensitive to proper compaction and drainage.

Construction sequence of a reinforced earth wall

1. Excavate, level and compact the foundation; cast unreinforced concrete **leveling pad** (≈ 150 mm × 300 mm).
2. Erect first row of facing panels, plumb and aligned; temporary wedges / bracing.
3. Place and compact backfill in 150–200 mm lifts up to the first reinforcement level (≥ 95 % MDD).

4. Lay reinforcement perpendicular to the face, flat, and connect it to the panels.
5. Spread the next lift; avoid heavy rollers within 1 m of the facing (use hand compactors).
6. Repeat layer-by-layer; provide **drainage** behind and below the reinforced zone; cast coping beam at top.

Formula sheet – Unit I

Quantity	Expression
Interface friction	$\mu = \tau/\sigma_n = \tan\delta$; $C_i = \tan\delta / \tan\varphi$
Pull-out resistance	$P_r = 2\, b\, L_e\, \mu^* \sigma_v$
Enhanced confinement	$\Delta\sigma_3 = T/(S_v S_h)$; $\Delta\sigma_1 = \Delta\sigma_3 K_p$
Pseudo-cohesion	$c_R = (\Delta\sigma_3/2)\sqrt{K_p}$; $K_p = \tan^2(45^\circ + \varphi/2)$
Rankine coefficient	$K_a = \tan^2(45^\circ - \varphi/2) = (1 - \sin\varphi)/(1 + \sin\varphi)$

Important questions

1. Explain the concept of reinforced earth. How is soil strengthened by reinforcement?
2. Explain equal confining pressure and pseudo-cohesion concepts with Mohr circles.
3. Define friction coefficient. Describe pull-out test and direct shear test.
4. List the factors affecting soil–reinforcement friction coefficient.
5. Write briefly on materials and applications of reinforced earth.

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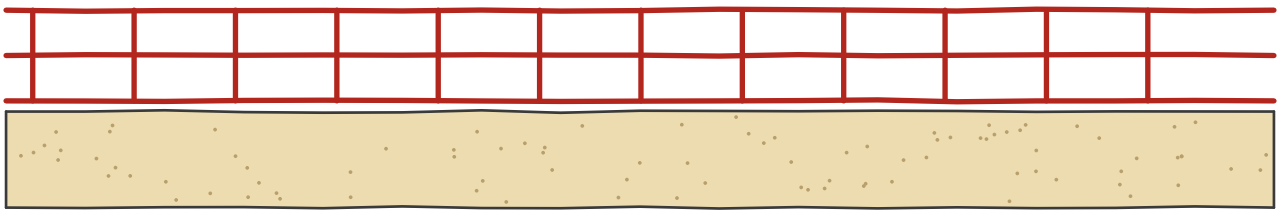
UNIT – II

Geosynthetics

Syllabus topics covered in these notes:

- Geosynthetics: advantages over conventional materials
- Classification based on material type and function; types of geosynthetics
- Functions of geosynthetics; tests on geosynthetics
- Durability aspects of geosynthetics; applications of geosynthetics

soil + reinforcement = strong composite



1. Geosynthetics – Introduction

Geosynthetics = geo (earth) + synthetic (man-made polymer). They are **planar, polymeric (synthetic or natural) products** manufactured for use with soil, rock or other geotechnical material as an integral part of a man-made project, structure or system (ASTM D4439).

Raw materials (polymers)

- **Polypropylene (PP)** – most geotextiles, geogrids; cheap, UV-sensitive, creeps.
- **Polyethylene (HDPE, LDPE, LLDPE)** – geomembranes, geogrids, geonets; chemically resistant.
- **Polyester (PET)** – high strength, low creep; geogrids, woven geotextiles; sensitive to high pH (hydrolysis).
- **Polyvinyl chloride (PVC)** – flexible geomembranes; **Polyamide (nylon)**, **aramid**, **PVA** – special reinforcement, **EPS** – geofoam.
- Additives: carbon black (2–3 %), antioxidants and UV stabilisers are added to improve durability.

Advantages over conventional materials

Conventional (sand, stone, steel, clay)	Geosynthetics
Natural resources, variable quality	Factory made – uniform, quality controlled
Thick layers, heavy haulage	Thin rolls, easy transport and handling
Slow, labour and equipment intensive	Fast installation, little equipment
Steel corrodes; clay layers crack	Corrosion free, flexible, chemically resistant
Large quarrying; high carbon cost	Saves aggregate and land; eco-friendly
High cost for long haul	Cost saving 20–60 % in many applications

Limitations

- Damaged by **UV light** if left exposed; sensitive to fire and some chemicals.
- **Creep** under sustained load; installation damage; long-term data limited (≈ 60 years of use).

- Needs correct design, specification and good site supervision.

2. Classification of Geosynthetics

Geosynthetics can be classified (i) by **type / product form**, (ii) by **polymer (material)** and (iii) by **function**. Main types are shown below.

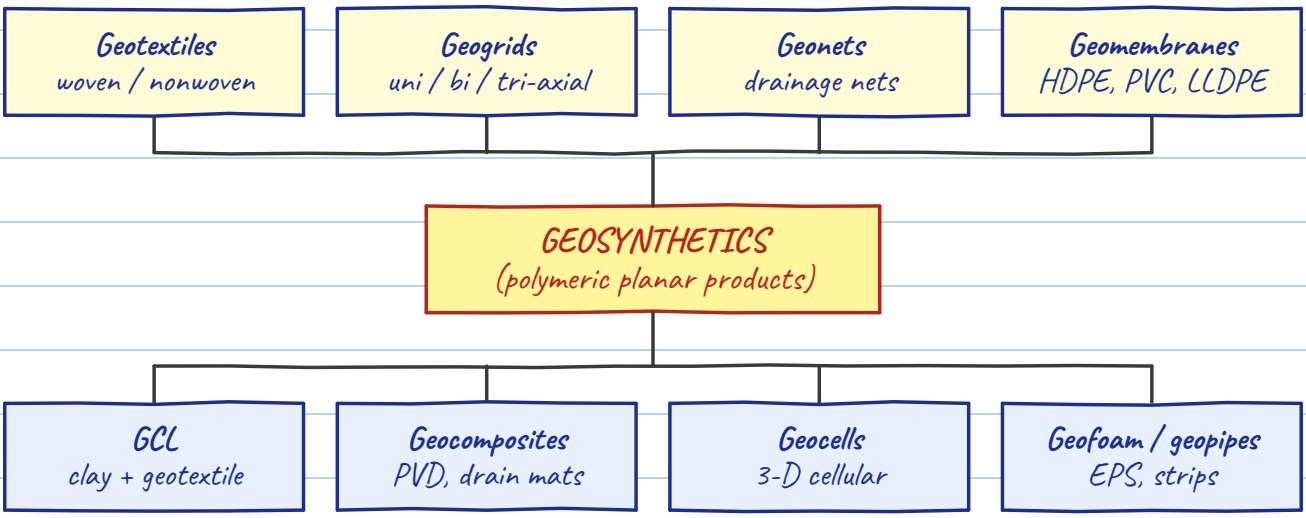


Fig 2.1 Types of geosynthetics

2.1 Types of geosynthetics

(i) **Geotextiles** – permeable, fabric-like sheets of PP/PET/PE fibres.

- **Woven** – slit-film or monofilament yarns interlaced at right angles; high strength, low strain, low permeability → reinforcement, separation.
- **Nonwoven** – fibres bonded randomly (needle-punched, heat-bonded, resin-bonded); thick, high elongation, high permeability → filtration, drainage, protection.
- **Knitted** – loops formed, used for special reinforcement and erosion control.

(ii) **Geogrids** – open grid-like sheets with large apertures (10–100 mm), stiff ribs; interlock with aggregate → **reinforcement**.

- **Uniaxial** (high strength in one direction – walls, slopes), **biaxial** (both directions – roads, platforms), **triaxial** (triangular apertures – better load spread).

- Manufactured by extrusion & stretching (HDPE/PP), weaving + coating (PET) or welding of strips.

(iii) Geonets – two sets of parallel ribs crossing at an angle; high in-plane flow → **drainage.**

(iv) Geomembranes – continuous impermeable sheets (HDPE, LLDPE, PVC, fPP, EPDM); $k \approx 10^{-11}$ to 10^{-13} cm/s → **barrier / containment**, thickness 0.5–2.5 mm.

(v) Geosynthetic clay liners (GCL) – sodium bentonite sandwiched between geotextiles; self-healing; $k \leq 10^{-9}$ m/s.

(vi) Geocomposites – combination of two or more types, e.g. geotextile + geonet drain, **prefabricated vertical drains (PVD / wick drains)**, geogrid + nonwoven.

(vii) Geocells (geoweb) – 3-D honeycomb cells filled with soil/aggregate → confinement, slope and base stabilisation.

(viii) Others – geofoam (EPS blocks, light fill), geopipes, geostrips/geobelts, geomats, erosion control blankets, geotubes / geobags.

2.2 Classification by polymer and by function

Product	Common polymer	Main function
Woven geotextile	PP, PET	Separation, reinforcement
Nonwoven geotextile	PP, PET	Filtration, drainage, protection
Geogrid	HDPE, PP, PET	Reinforcement
Geonet	HDPE	Drainage
Geomembrane	HDPE, PVC, LLDPE	Containment (barrier)
GCL	Bentonite + PP	Containment
Geocomposite / PVD	PP + PE core	Drainage, filtration
Geocell	HDPE	Reinforcement, confinement

Manufacturing in brief

- **Woven GT:** polymer → yarns (slit-film / monofilament) → woven on looms.

- **Nonwoven GT:** filaments extruded → random web → bonded by **needle punching** (barbed needles entangle fibres), **heat bonding** or resin.
- **Geogrid:** extruded sheet punched with holes and **stretched** (uni- or biaxially) to align polymer chains; or PET yarns knitted/woven and PVC/bitumen coated.
- **Geomembrane:** flat-die / blown-film extrusion or calendering; panels joined at site by **thermal (hot wedge) fusion** or **extrusion welding**; seams tested by peel, shear and air-channel tests.
- **GCL:** bentonite layer needle-punched or stitched between two geotextiles; swells on hydration and seals punctures.

Prefabricated vertical drain (PVD) – geocomposite

Band drain about 100 mm × 4 mm: plastic **grooved core** wrapped by a **filter geotextile**, pushed into soft clay with a mandrel at 1–2.5 m spacing. It shortens the drainage path from the clay thickness to **half the drain spacing**, so consolidation takes months instead of years.

$$t = (D^2 / 8c_h) \cdot [\ln(D/d_w) - 0.75] \cdot \ln[1/(1 - U)]$$

$D = 1.05\ s$ (triangular) or $1.13\ s$ (square pattern), $d_w = 2(a + b)/\pi$ = equivalent drain diameter, c_h = horizontal coefficient of consolidation, U = degree of consolidation (Barron; smear ignored).

Example (PVD)

$s = 1.5$ m triangular → $D = 1.575$ m; $a = 100$ mm, $b = 4$ mm → $d_w = 66$ mm; $c_h = 3\ m^2/yr$; $U = 90\ \%$

$t = (1.575^2/(8\times3)) \times [\ln(23.9) - 0.75] \times \ln(10) = 0.1034 \times 2.42 \times 2.303 =$ 0.58 year (≈ 7 months)

3. Functions of Geosynthetics

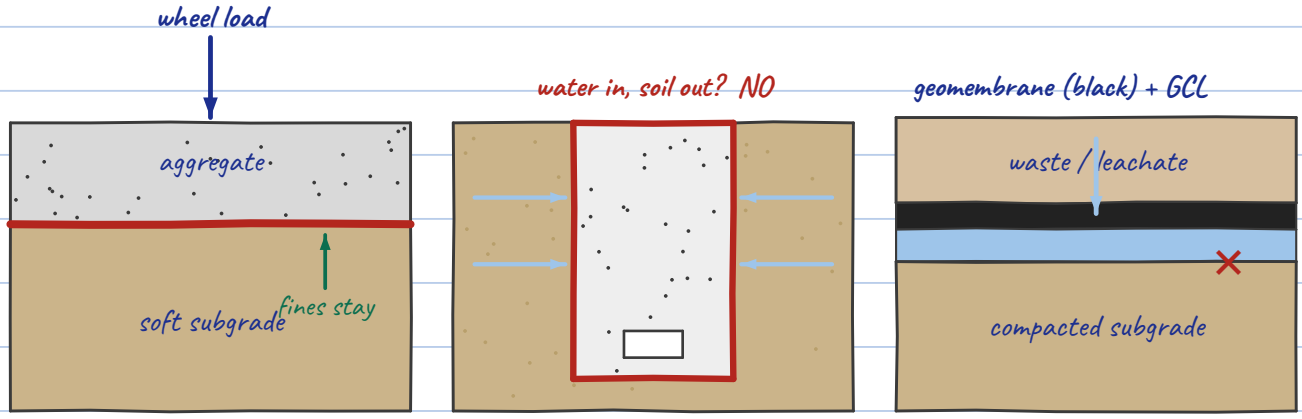


Fig 2.2 Separation, filtration/drainage and containment

1. Separation – keeps two dissimilar materials (soft subgrade and aggregate base) from mixing. Without it, fines pump up and stone punches down, so the base thickness is lost. Properties: tensile strength, puncture, burst, apparent opening size. Saves aggregate (≈ 30–50 %).

2. Reinforcement – adds **tensile resistance** to soil (geogrid, woven geotextile, geostrip): retaining walls, steep slopes, embankments on soft soil, bearing capacity improvement. Properties: tensile strength and modulus, creep, soil-interaction (pull-out).

- **Lateral restraint** – tensile reinforcement holds soil/aggregate from spreading sideways (see confining pressure concept).
- **Bearing-capacity (failure-plane) mechanism** – forces the failure surface to a deeper, stronger zone.
- **Tensioned-membrane effect** – after rutting, curved sheet carries part of the wheel load by vertical component of tension.

3. Filtration – allows water to pass but retains soil particles (like a sand filter). Typical use: behind retaining walls, around drains, under riprap.

Filter criteria (geotextile)

Retention: $O_{95} \le B \cdot D_{85}$ ($B = 1$ for $C_u \le 2$ or ≥ 8; $B = 0.5\,C_u$ for $2 < C_u < 4$; $B = 8/C_u$ for $4 \le C_u \le 8$)

Permeability: $k_g \ge k_s$ (critical: $\ge 10\,k_s$); permittivity $\psi \ge 0.5\ s^{-1}$ (fines < 15 %), 0.2 (15–50 %), 0.1 (> 50 %)

Anti-clogging: O_{95} not too small; nonwoven preferred for fine soils.

4. Drainage (in-plane flow) – carries water/gas within its plane (geonet, thick nonwoven, geocomposite). Transmissivity $\theta = k_p \cdot t$ (m^2/s); flow $q = k_p\, i\, t\, W$. Used in landfill leachate collection, behind walls, under roads, PVDs for consolidation of soft clay.

5. Containment (barrier) – geomembrane / GCL stop liquid or gas movement: landfill liners and caps, canal and pond lining, mining heap-leach pads.

6. Protection – cushion layer (thick nonwoven) protects a geomembrane from puncture by stones; **7. Erosion control** – mats and blankets hold soil on slopes and channels until vegetation grows.

Example 1

Example (filter): soil $D_{85} = 0.4$ mm, $C_u = 5 \rightarrow B = 8/5 = 1.6$

$O_{95} \le 1.6 \times 0.4 = 0.64$ mm (geotextile AOS must be smaller than this).

4. Tests on Geosynthetics

Properties are tested in the laboratory as per **ASTM / ISO / IS 13162** (Indian standard series for geotextiles).

4.1 Physical properties

- **Mass per unit area** (g/m^2) – ASTM D5261; **thickness** (mm) at 2 kPa – ASTM D5199; **specific gravity / density** of polymer; **roll size**; **stiffness**.

4.2 Mechanical properties

- **Wide-width tensile test** (ASTM D4595 / ISO 10319): specimen 200 mm wide × 100 mm gauge, strain rate 10 %/min. Gives T_{ult} (kN/m), strain at failure, secant modulus $J = T/\varepsilon$ (at 2 %, 5 %).

- **Grab / strip tensile** (ASTM D4632) – index test, small specimen held by 25 mm jaws.
- **Trapezoidal tear** (D4533), **puncture** (CBR puncture D6241, drop-cone), **burst** (D3786).
- **Creep test** (D5262, tension creep) – constant load held for 1000–10 000 h.
- **Seam / joint strength**; **interface friction** (direct shear, pull-out, inclined plane – see Unit I).

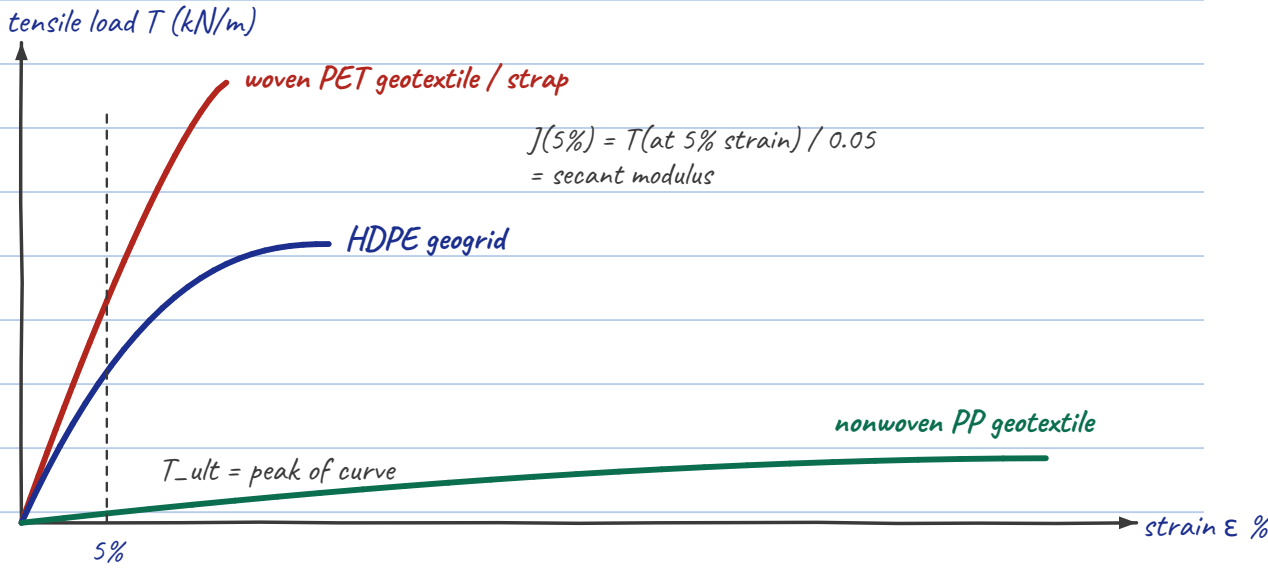


Fig 2.3 Load–strain behaviour of geosynthetics

4.3 Hydraulic properties

- **Permittivity / permeability** (ASTM D4491; constant or falling head) – cross-plane flow.
- **Apparent opening size, AOS (O_{95})** (ASTM D4751) – dry sieving with glass beads.
- **Transmissivity** (ASTM D4716) – in-plane flow under load and gradient.
- **Gradient ratio / clogging** (ASTM D5101) – long-term filter compatibility with soil.

Examples

Wide-width test: specimen 200 mm wide; peak load 12 kN → T_{ult} = 12/0.2 = 60 kN/m.

Load at 5 % strain = 6 kN → $T_{5\%}$ = 30 kN/m → J = 30/0.05 = 600 kN/m.

Permittivity $\psi = Q / (A \cdot \Delta h)$ $[s^{-1}]$; $k_n = \psi \cdot t$. If $\psi = 0.8\ s^{-1}$, t = 2 mm → k = 0.8 × 0.002 = 1.6×10^{-3} m/s.

4.4 Endurance and durability tests

- **UV resistance** (ASTM D4355) – xenon-arc weatherometer, 500 h; retained strength reported.
- **Chemical, oxidation and hydrolysis** (immersion in solution at high temperature), **biological** resistance (soil burial).
- **Installation damage** (ASTM D5818) – field compaction trial with aggregate, then retained strength.
- **Abrasion** (D4886), **temperature** effects, **stress cracking** (geomembranes, D5397).

Summary of tests

Property	Test (ASTM)	Result / unit
Mass per unit area	D5261	g/m^2
Thickness	D5199	mm at 2 kPa
Tensile (wide width)	D4595	kN/m, % strain, J
Puncture / burst	D6241 / D3786	N / kPa
Permittivity	D4491	s^{-1}
Apparent opening size	D4751	O95 (mm)
Transmissivity	D4716	m^2/s
UV resistance	D4355	% strength retained

5. Durability Aspects

Design life of geosynthetics (60–120 years for permanent works) depends on resistance to the following degradation mechanisms:

Mechanism	Material mostly affected	Remedy
Creep / stress rupture	PP, HDPE (PET low)	Use creep factor; choose PET
Installation damage	All (esp. nonwoven, thin)	Cover quickly, select fill
UV degradation	PP, PE (without carbon black)	Cover within days / weeks
Oxidation (thermal)	PP, PE	Antioxidants, cooler soil
Hydrolysis	PET at pH > 9 and warm	Check soil pH, use PP/HDPE in lime soil
Chemical attack / stress crack	HDPE membranes, PP	Compatibility test
Biological, clogging	Filters, drains	Correct AOS, gradient ratio

Allowable (long-term design) strength

$$T_{al} = T_{ult} / (RF_{ID} \times RF_{CR} \times RF_{D} \times FS)$$

RF_{ID} = installation damage (1.05–3.0), RF_{CR} = creep (PET 1.6–2.5; PP 4–5; HDPE 2.6–5), RF_{D} = chemical/biological durability (1.1–2.0), FS = overall factor of safety (often incl. uncertainty).

Example 2

T_{ult} = 60 kN/m, RF_{ID} = 1.1, RF_{CR} = 2.0, RF_{D} = 1.1, (FS applied separately)

T_{al} = 60 / (1.1 × 2.0 × 1.1) = 60 / 2.42 = 24.8 kN/m

6. Applications of Geosynthetics

Area	Geosynthetic used	Function
Roads, railways, airfields	Geotextile, geogrid, geocell	Separation, reinforcement
Retaining walls, steep slopes	Geogrid, geostrip, woven GT	Reinforcement
Embankments on soft soil	High strength GT / geogrid, PVD	Reinforcement, drainage
Landfills, ponds, canals	Geomembrane, GCL, geonet	Containment, drainage
Drains, dams, subsurface	Nonwoven GT, geocomposite	Filtration, drainage
Coastal / river works	GT tubes, mats, filters	Erosion control, filtration
Tunnels, mining	Membrane, shotcrete liner, GT	Waterproofing, protection

Typical landfill liner system (bottom to top)

1. Compacted subgrade → **compacted clay / GCL** (primary barrier).
2. **HDPE geomembrane** (1.5–2 mm) – leachate barrier.
3. **Nonwoven geotextile cushion** – protects the membrane from stones.
4. **Drainage layer** (gravel or geonet) with perforated leachate pipes.
5. **Filter geotextile** – keeps waste fines out of the drainage layer, then waste.

Selection of a geosynthetic

- Identify the **main function** first (separation, reinforcement, filtration, drainage, containment).
- Check required **properties** (strength, AOS, permittivity) against site soil and loads.
- Check **durability** (pH, temperature, UV, chemicals), **installation survivability** and **cost**.

Important questions

1. Define geosynthetics. List their advantages over conventional materials.
2. Classify geosynthetics by type, material and function; explain geotextiles and geogrids.

3. Explain the functions of geosynthetics with sketches and examples.
4. Describe the wide-width tensile test and permittivity test.
5. Discuss the durability aspects of geosynthetics; explain reduction factors.

HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

Subject Code : 23A01702a

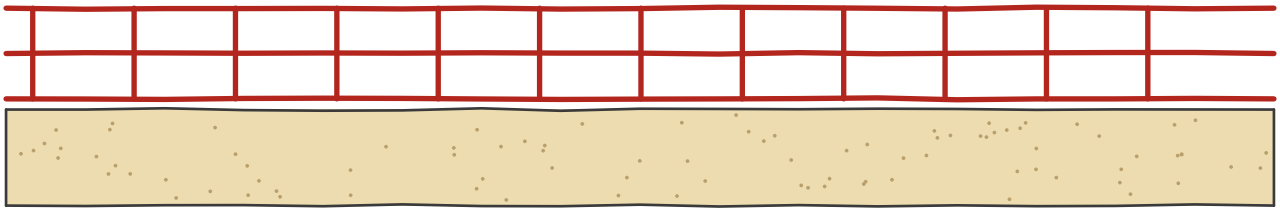
UNIT – III

Reinforced Earth Retaining Walls

Syllabus topics covered in these notes:

- Reinforced earth retaining walls: introduction, stability mechanisms
- Design of RE wall: selection of materials, geotechnical analysis
- Reinforcement layout and spacing; stability analysis
- Advantages over conventional retaining walls

soil + reinforcement = strong composite



1. Reinforced Earth Retaining Walls

A **reinforced earth (RE) / mechanically stabilised earth (MSE) wall** is a retaining structure in which a mass of compacted granular fill is reinforced by horizontal layers of strips, grids or sheets that are connected to a **facing**. The reinforced mass acts like a **gravity wall** that resists the thrust of the retained soil behind it.

Parts of an RE wall

- **Reinforced zone** – compacted backfill + reinforcement (length L, height H).
- **Retained fill / natural soil** behind the reinforced zone.
- **Facing** (precast panels, blocks, wrap-around, welded wire) and **leveling pad**.
- **Drainage** – behind and below the wall (weep holes, geocomposite, sub-surface drain).
- **Foundation soil** – must have adequate bearing capacity and limited settlement.

Types

Type	Reinforcement	Facing
Reinforced Earth (Vidal)	Galvanised steel strips	Cruciform RCC panels
Welded wire / bar-mat wall	Steel grids	Wire basket, panels
Geosynthetic wall	Geogrid / geotextile / geostrip	Blocks, wrap-around, panels
Soil nailing / anchored (different)	Passive nails	Shotcrete

2. Stability Mechanisms

Two classical ways of looking at an RE wall:

- **Tie-back wedge method** – the active wedge (Rankine/Coulomb) behind the facing tries to slide; reinforcement beyond the wedge acts as a **tie-back** (anchor). Used for **extensible** reinforcement.

- **Coherent gravity method** – the reinforced block behaves as a rigid, coherent mass; the line of maximum tension is **not** the Rankine line but a bilinear curve. Used for **inextensible** (steel) reinforcement.

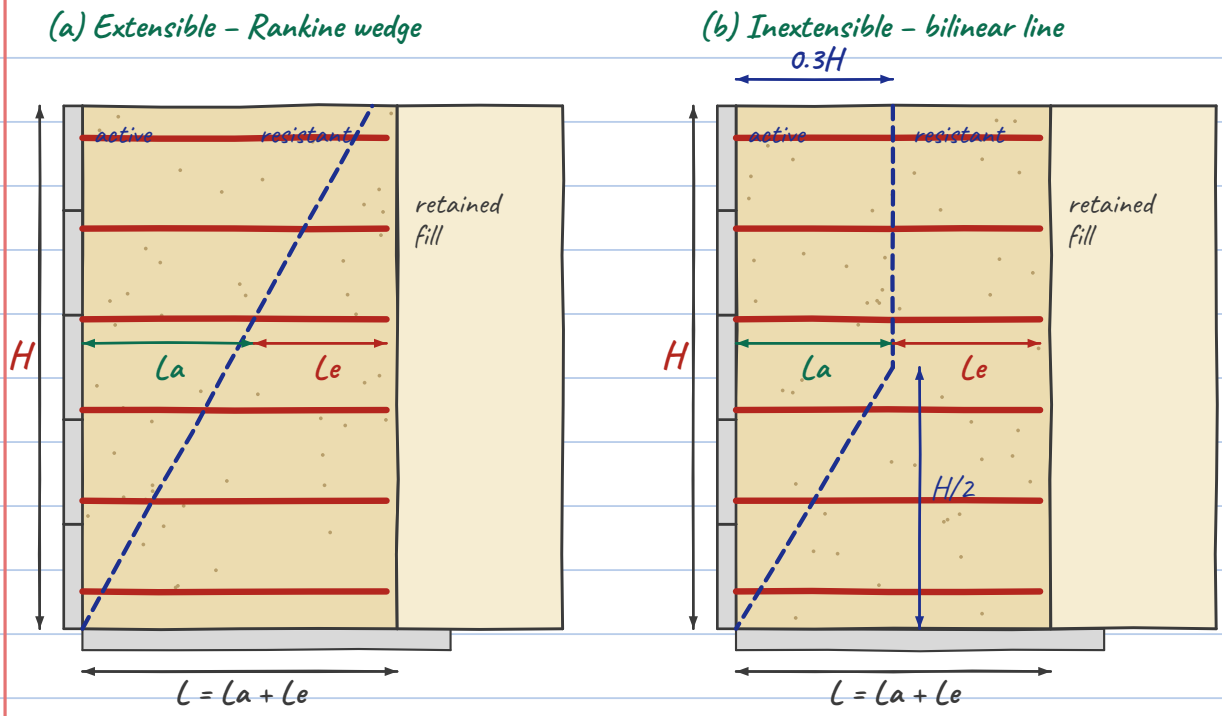


Fig 3.1 Failure surfaces and reinforcement lengths

(a) Rankine plane: $L_a = (H - z)\tan(45° - \varphi/2)$

(b) Bilinear: vertical 0.3H from face down to H/2, then straight to the toe

Mechanism

Active zone (La) → soil tries to move out and pulls the strip ⇒ tension T.

Resistant zone (Le) → soil holds the strip by friction ⇒ pull-out resistance.

Total length L = La + Le at every level.

Lateral earth pressure coefficient in the reinforced zone

Reinforcement	K at top (z = 0)	K at depth ≥ 6 m
Steel strips (original Terre Armée)	K0 = 1 – sin φ	Ka
Steel strips (AASHTO)	1.7 Ka	1.2 Ka
Steel grids / bar mats	≈ 2.5 Ka	1.2 Ka
Geosynthetics (extensible)	Ka	Ka

Reason: inextensible reinforcement prevents lateral strain so the soil stays close to the **at-rest** state near the top, where compaction-induced stresses are large; extensible sheets allow the soil to reach the active state.

Tie-back wedge vs coherent gravity

Feature	Tie-back wedge	Coherent gravity
Reinforcement	Extensible (geosynthetic)	Inextensible (steel)
Failure surface	Rankine plane $45° + \varphi/2$	Bilinear max-tension line
Pressure coefficient	Ka	K0 → Ka (1.7→1.2 Ka)
Idea	Strips act as anchors	Block acts as gravity wall
Used in	FHWA geosynthetic design	AASHTO / Reinforced Earth

3. Design of RE Wall – Steps

1. **Geometry** – wall height H, surcharge q, slopes, ground water, design life (75–100 years).
2. **Soil parameters** – reinforced fill (φ_r, γ_r), retained fill (φ_b, γ_b), foundation soil (c, φ, q_a).
3. **Select materials** (below).
4. **Trial** reinforcement length $L \geq 0.7H$ (min 2.5 m), spacing S_v, type.
5. **External stability** – sliding, overturning, bearing, global/ settlement.
6. **Internal stability** – rupture and pull-out at each layer; connection check.
7. **Facing / drainage details**, then revise and finalise.

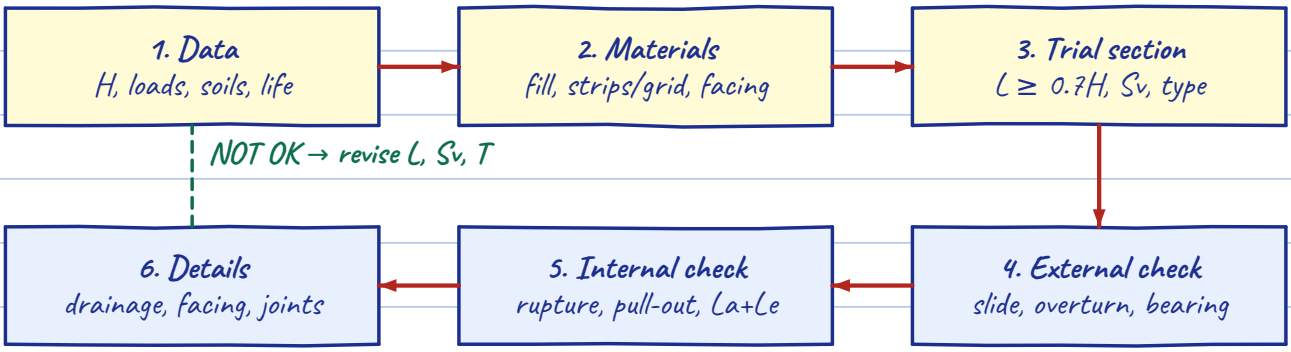


Fig 3.3 Design flow of an RE wall

3.1 Selection of materials

- **Fill:** granular, free-draining (fines ≤ 15 %, PI ≤ 6, max size ≤ 75 mm), $\varphi \geq 30°–34°$, compact to ≥ 95 % MDD. Electrochemical limits if steel is used.
- **Reinforcement:** choose by life, environment and cost: galvanised steel (strength, no creep) vs geogrid / geostrip (corrosion free). Use long-term strength $T_{al} = T_{ult}/(RF_{ID}\,RF_{CR}\,RF_{D})$.
- **Facing:** RCC panels (M30+), modular blocks, gabions, wrap-around, welded wire with vegetation – choose for height, aesthetics, tolerance to settlement.
- **Drainage:** granular blanket or geocomposite behind the reinforced zone; weep holes; toe drain; avoid water behind the wall.

3.2 Geotechnical analysis – external stability

The reinforced block (width L, height H) is treated as a **rigid gravity wall** acted on by the thrust of the retained soil.

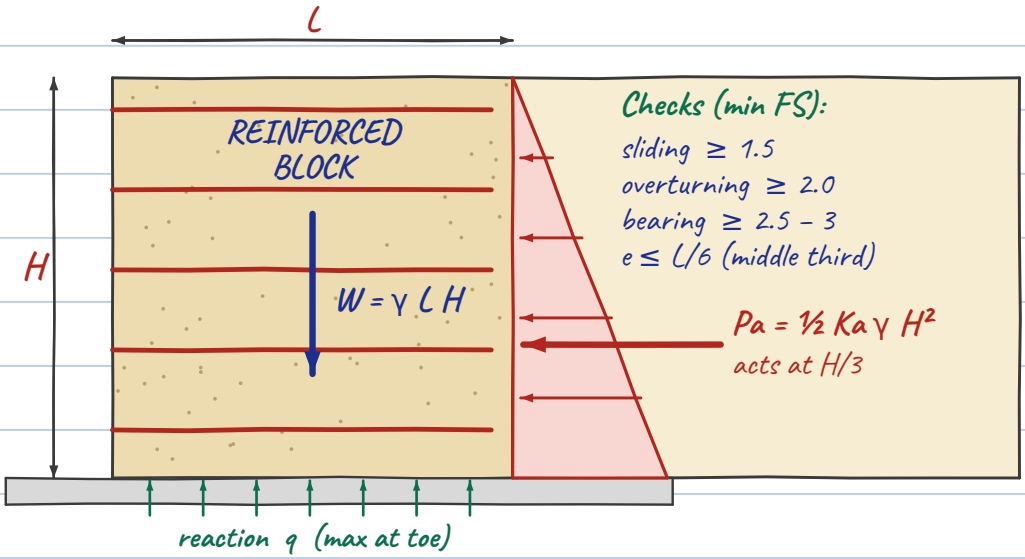


Fig 3.2 External stability of the reinforced block

$$P_a = \tfrac{1}{2} K_a \gamma_b H^2 \ (+ K_a q H \text{ for surcharge } q)$$

$$W = \gamma_r L H$$

- **Sliding:** $FS_s = W \tan \varphi_{min} / P_a \geq 1.5$ (φ_{min} = least of φ_r, $\varphi_{foundation}$, interface).
- **Overturning** about toe: $FS_o = M_R / M_O = (W \cdot L/2) / (P_a \cdot H/3) \geq 2.0$.
- **Eccentricity** $e = L/2 - (M_R - M_O)/W \leq L/6$.
- **Bearing:** $q_{max} = W / (L - 2e) \leq q_{all}$ ($FS \geq 2.5$–3 on $q_{ult} = cN_c + \tfrac{1}{2}\gamma (L - 2e) N_\gamma + \gamma D N_q$).
- **Global / deep-seated stability:** Bishop or Janbu method with reinforced block as part of slope, $FS \geq 1.3$–1.5. **Settlement:** total and differential (consolidation of foundation).

3.3 Internal stability

At depth z, the **horizontal stress** and the **tension** to be resisted by each reinforcement layer (vertical spacing S_v):

$$\sigma_h = K (\gamma z + q)$$

$$T_{max} = \sigma_h S_v \text{ (per metre run of wall for sheets / grids)}$$

$$T_{max} = \sigma_h S_v S_h \text{ (per strip, horizontal spacing } S_h)$$

- **Rupture check:** $T_{al} \geq T_{max}$ (using $FS \geq 1.5$ on allowable strength).

- **Pull-out check**: required embedment $L_e \geq FS \cdot T_{max} / (2\, b\, F^* \sigma_v\, C)$, $FS \geq 1.5$, min $L_e = 1$ m.
- Total length at each level $L = L_a + L_e$; adopt the greatest required, $L \geq 0.7H$ (and ≥ 2.5 m). Longer bottom layers sometimes used for sliding.
- **Connection** to facing should carry T_{max} (top layers) – check strength of connection and facing.

3.4 Reinforcement layout and spacing

- **Vertical spacing S_v**: 0.3–0.8 m (strips ≈ 0.75 m; geogrid 0.3–0.6 m; wrap-around geotextile 0.3–0.6 m). Smaller spacing → smaller tension per layer and better local stability.
- **Horizontal spacing S_h** (strips): 0.75–1.5 m, matched to panel width; continuous sheets have 100 % coverage.
- First layer at **0.15–0.3 m** above the base; last layer within 0.3 m of the top (below coping).
- Reinforcement laid **horizontally and perpendicular** to the wall face; avoid kinks; no overlap of strips in plan; sheets lapped ≥ 0.3 m.
- Layers may be **lengthened at the top** where surcharge / traffic loads act; length is usually **uniform** with height.

Example 1 – internal

Worked example. $H = 6$ m, $\gamma = 18$ kN/m^3, $\varphi = 32°$ (reinforced and retained fill), geogrid, $S_v = 0.6$ m, no surcharge.

$K_a = (1 - \sin 32°)/(1 + \sin 32°) = 0.307$

Layer at $z = 3$ m: $\sigma_v = 54$ kPa, $\sigma_h = 0.307 \times 54 = 16.6$ kPa → $T = 16.6 \times 0.6 = 9.95$ kN/m

Bottom layer $z = 5.7$ m: $\sigma_v = 102.6$, $\sigma_h = 31.5$ → $T = 18.9$ kN/m → $T_{al} \geq 1.5 \times 18.9 = 28.4$ kN/m

Example 1 – length

Pull-out at z = 3 m: $F^* = (2/3)\tan 32° = 0.417$; $L_e = 1.5 \times 9.95 / (2 \times 0.417 \times 54) = 0.33$ m → adopt 1.0 m (minimum).

$L_a = (6 - 3)\tan 29° = 1.66$ m → $L = 1.66 + 1.0 = 2.7$ m; but $L \geq 0.7H = 4.2$ m (governs) → adopt $L = 4.2$ m.

Example 1 – external

External ($L = 4.2$ m): $W = 18 \times 4.2 \times 6 = 453.6$ kN/m; $P_a = ½ \times 0.307 \times 18 \times 6^2 = 99.6$ kN/m

$FS_{sliding} = 453.6 \times \tan 32° / 99.6 = 283.5/99.6 = 2.85 > 1.5$ OK

$M_R = 453.6 \times 2.1 = 952.6$; $M_O = 99.6 \times 2 = 199.2$ → $FS_o = 4.78 > 2$ OK

$e = 2.1 - (952.6 - 199.2)/453.6 = 0.44$ m $< L/6 = 0.70$ m OK; $q_{max} = 453.6/(4.2 - 0.88) = 136.5$ kPa $\leq q_{all}$ (check)

3.5 Surcharge, live load and seismic effects

- **Uniform surcharge q** (traffic ≈ 10–12 kPa equivalent): increases σ_v and $\sigma_h = K q$ – add to external thrust $K_a q H$ and to tension at every layer.
- **Strip / footing loads** near the wall: spread at 2V : 1H (or Boussinesq) to find extra σ_v and σ_h at each level.
- **Seismic:** pseudo-static (Mononobe–Okabe) – add dynamic thrust ΔP_{AE} on the retained zone and inertia force on the reinforced block; pull-out resistance checked behind the (enlarged) active zone; use longer layers ($L \geq 0.8H$) in high seismic zones.
- **Water:** hydrostatic pressure behind the wall must be avoided by drainage; if unavoidable, use buoyant unit weight and add water thrust.

Example 2 – strength

Example 2 – steel strips. $H = 8$ m, $\varphi = 34°$, $\gamma = 18$, strips 50 × 4 mm galvanised, $S_v = S_h = 0.75$ m, life 75 years.

Corrosion: zinc (85 µm) lasts ≈ 2 yr × 15 + 55/4 ≈ 16 yr; steel loss ≈ 12 µm/yr × 59 yr ≈ 0.71 mm per face → $t_c = 4 - 1.42 = 2.58$ mm

$A_c = 50 \times 2.58 = 129$ mm²; allowable stress $0.55 f_y = 0.55 \times 450 = 247$ MPa → $T_{al} = 31.9$ kN/strip

Example 2 – length

Bottom layer $z = 7.5$ m: $K_a = 0.283$, $K = 1.2 K_a = 0.339$; $\sigma_v = 135$ kPa → $\sigma_h = 45.8$ kPa

$T_{max} = 45.8 \times 0.75 \times 0.75 = 25.8$ kN < 31.9 OK

Pull-out: $F^* = \tan 34° = 0.675$ → $L_e = 1.5 \times 25.8 / (2 \times 0.05 \times 0.675 \times 135) = 4.25$ m

L_a (bilinear) at $z = 7.5$ m = 0.3 m → $L = 4.55$ m; but $L \geq 0.7H = 5.6$ m → adopt $L = 6$ m (rounded).

Example 3

Example 3 – effect of surcharge $q = 12$ kPa on retained fill (Example 1 wall, $L = 4.2$ m):

Extra thrust = $K_a q H = 0.307 \times 12 \times 6 = 22.1$ kN/m acting at $H/2 = 3$ m

$P_{a,total} = 99.6 + 22.1 = 121.7$; $FS_s = 283.5/121.7 = 2.33$ OK

$M_O = 199.2 + 22.1 \times 3 = 265.5$; $FS_o = 952.6/265.5 = 3.59$ OK

Top layer ($z = 0.3$ m): $\sigma_h = 0.307 (18 \times 0.3 + 12) = 5.3$ kPa → $T = 5.3 \times 0.6 = 3.2$ kN/m (higher than without q).

3.6 Facing and connections

- Facing does **not** carry earth pressure – it only retains the face soil between layers and resists local erosion; connection load is ≈ 50–100 % of T_{max} at the top and reduces to ≈ 50–60 % near the base (extensible).
- Precast panels (≈ 1.5 m × 1.5 m) have **joints** with filler and filter strip so that fines do not wash out and differential settlement is accommodated.

- Connections must be corrosion protected (steel) and have creep / strength reduction tested (geosynthetic, block–grid connection).
- Wall **batter** and construction tolerance: ≈ 1° backward batter; alignment ± 20–25 mm.

Short answers

- L_a: length of reinforcement in the active zone (not mobilising resistance). L_e: length in the resistant zone (anchorage length).
- **Why $L \geq 0.7H$?** To provide enough width for the reinforced block to resist sliding, overturning and bearing as a gravity wall.
- **Why different K for steel and geosynthetics?** Inextensible steel restrains lateral strain → higher K at shallow depth.
- **Drainage importance:** a water-filled backfill adds hydrostatic thrust and reduces friction.

4. Construction and Details

1. Prepare and compact the foundation; cast **leveling pad**; set the first facing course plumb.
2. Place reinforced fill in **150–200 mm** lifts; compact ≥ 95 % MDD; light equipment within 1 m of the face.
3. Lay reinforcement flat, pull taut, connect to the facing; fill from the face side moving away.
4. Install drainage and filter geotextile behind the face and at the back of the reinforced zone.
5. Repeat; finish with coping beam, crash barrier, surface drainage and pavement.

- Provide **joints** (compressible filler) in panels to allow settlement; **batter** wall ≈ 1° backward to counter outward movement.
- Control **construction tolerances** – alignment ± 20–25 mm, verticality 1:100.

Common causes of distress

1. **Poor drainage** – water pressure, softened fill (most common cause).
2. **Poor compaction** or use of fine / clayey fill → bulging of the face.
3. **Too short or too weak reinforcement**; corrosion / creep / damage of reinforcement.
4. **Foundation settlement** or bearing failure; scour at the toe (river / sea walls).
5. **Overloading** by unplanned surcharge or construction traffic near the face.

Monitoring: survey targets on facing (outward movement < 1 % H), settlement markers, strain gauges on reinforcement, piezometers and inspection of weep holes.

5. Advantages over Conventional Retaining Walls

Point	Conventional (gravity / cantilever)	Reinforced earth wall
Foundation	Needs strong, deep foundation	Flexible; works on weaker soil
Settlement	Rigid – cracks, tilts	Tolerates large differential settlement
Height	Economical up to ≈ 6–8 m	Economical to 10–30 m
Construction	Formwork, curing – slow	Prefab panels, layer-wise – rapid
Cost	Higher (concrete, steel)	25–50 % cheaper (typical)
Seismic	Rigid, brittle failure	Flexible, energy absorbing
Space / aesthetics	Larger base, one finish	Wide variety of facings

Limitations

- Needs **granular fill** and enough space behind the wall ($L \geq 0.7H$).
- Not suitable where the backfill zone is submerged or contaminated, or for walls with large utilities.
- Corrosion / creep / UV need checking; quality of compaction and drainage is critical.

Formula sheet – Unit III

Item	Expression
Rankine coefficient	$K_a = \tan^2(45^\circ - \varphi/2)$
Active thrust	$P_a = \tfrac{1}{2} K_a \gamma H^2 + K_a q H$
Tension in layer	$T = K_a \sigma_v S_v$ (sheets)
Pull-out length	$L_e \geq FS\, T / (2 b F^* \sigma_v)$
Sliding / overturning	$W \tan\varphi / P_a \geq 1.5$; $M_R / M_O \geq 2$
Eccentricity	$e = L/2 - (M_R - M_O)/W \leq L/6$

Important questions

1. Explain the mechanism of stability of a reinforced earth retaining wall.
2. Describe the design steps of an RE wall; discuss selection of materials.
3. Explain internal and external stability analysis with neat sketches.
4. Explain reinforcement layout and spacing. Compare RE walls with conventional walls.
5. Design an RE wall of 6 m height (given soil data) – numerical problem.

HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

Subject Code : 23A01702a

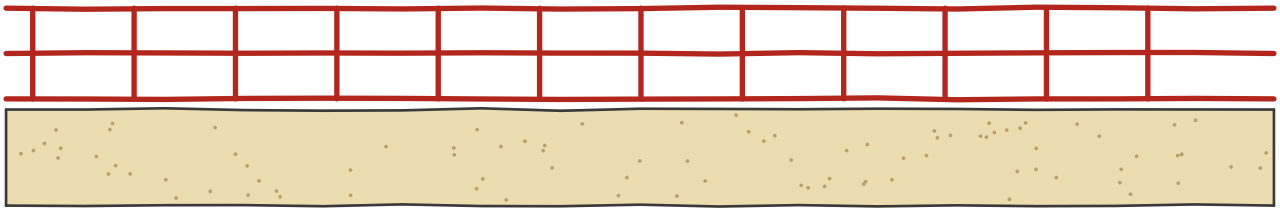
UNIT – IV

Reinforced Embankments

Syllabus topics covered in these notes:

- Reinforced embankments: introduction, design of reinforced embankment
- Foundation mattress below the embankment: purpose, function, components
- Design of reinforced mattress: settlement control, bearing capacity, long-term performance
- Field implementation and monitoring techniques

soil + reinforcement = strong composite



1. Reinforced Embankments – Introduction

Embankments for highways, railways, flood banks and approaches are often built on **soft, compressible clays** (low undrained strength c_u, high compressibility). An unreinforced embankment can fail by slip, spreading or bearing failure, or settle excessively. **Basal reinforcement** (high-strength geotextile / geogrid) laid at the base of the fill – the **foundation mattress** – raises stability and allows faster, higher construction.

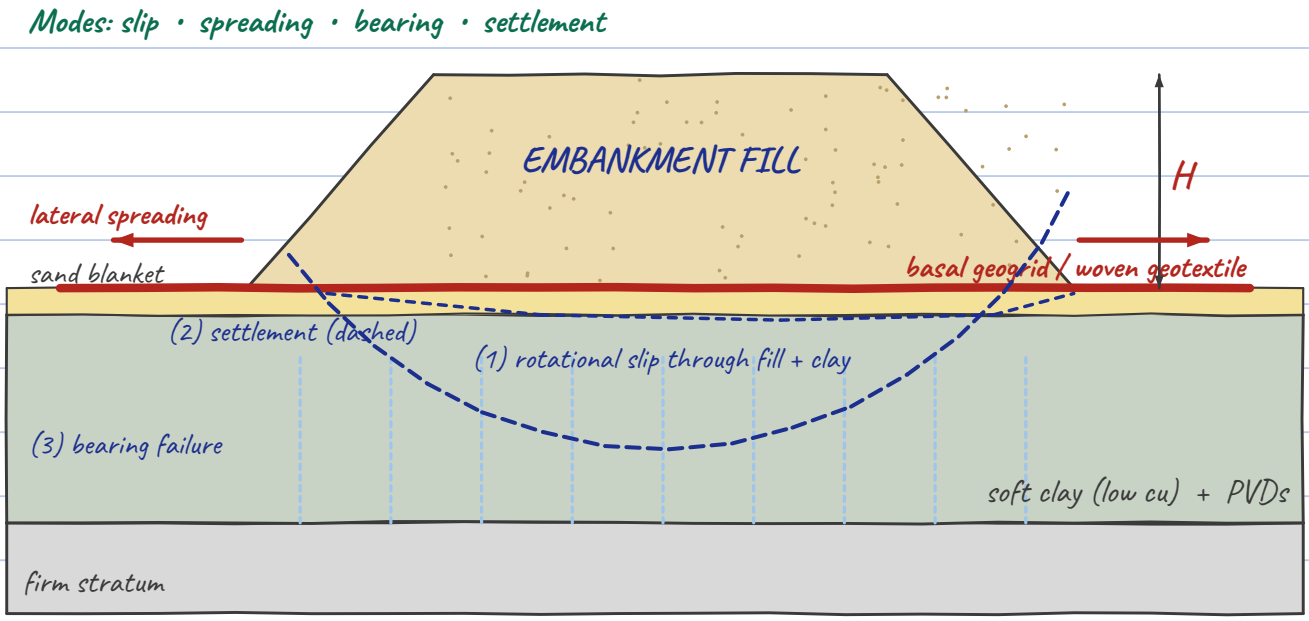


Fig 4.1 Embankment on soft ground with basal reinforcement

Failure modes to be checked

1. **Rotational (slope) failure** through fill and soft foundation – needs reinforcement tension T to resist moment.
2. **Lateral spreading / sliding** of the fill over the base – outward thrust $P_a = \frac{1}{2} K_a \gamma H^2$ resisted by tension and friction.
3. **Foundation (bearing) failure** – squeezing of soft clay: $q_{ult} \approx 5.14\, c_u$.
4. **Excessive total / differential settlement** and long-term consolidation.
5. **Reinforcement rupture, pull-out, or excessive strain** (> 5–6 %).

Types of reinforced embankment

- **Basal-reinforced embankment** on soft soil (geotextile / geogrid at the base).
- **Reinforced steep side-slope embankment** (reinforced soil slope, layers every 0.3–0.6 m to angles up to 70°) to save land.
- **Piled embankment** with geosynthetic platform bridging piles / stone columns.
- **Staged construction** with PVDs and preloading (reinforcement + drains).

2. Design of Reinforced Embankment

2.1 Data and requirements

- Embankment height H, side slope, fill properties (γ, φ), traffic surcharge q.
- Foundation soil: c_u profile, thickness of soft layer, C_c, e_0, c_v, c_h, water table.
- Required FS: slope stability ≥ 1.3 (short term) / 1.5 (long term); spreading ≥ 1.5; bearing ≥ 1.3–1.5.

2.2 Unreinforced critical height

$$H_c \approx 5.14\, c_u / \gamma$$

Example 1

c_u = 15 kPa, γ_{fill} = 19 kN/m³ → H_c = 5.14 × 15 / 19 = 4.06 m; safe height for FS = 1.3: ≈ 3.1 m.

If the embankment must be 6 m high → reinforcement / staging / drains are needed.

2.3 Stability against lateral spreading

Outward thrust of the fill (active) is resisted by the **tension in the basal geosynthetic**:

$$T_{req} \geq FS \cdot \tfrac{1}{2} K_a \gamma H^2, \quad K_a = \tan^2(45° - \varphi/2)$$

Also check **sliding of fill over reinforcement**: $FS = (\gamma H L \tan\delta) / (\tfrac{1}{2} K_a \gamma H^2) \geq 1.5$ and **pull-out / anchorage** of the geosynthetic beyond the failure zone.

Example 2

$H = 5$ m, $\gamma = 19$, $\varphi = 30°$ ($K_a = 0.333$): $P_a = ½ \times 0.333 \times 19 \times 25 = 79$ kN/m

$T_{req} = 1.5 \times 79 = 118$ kN/m (long-term allowable) → $T_{ult} = 118 \times (RF_{CR} \cdot RF_{ID} \cdot RF_D \approx 2.4) \approx 285$ kN/m → use PET geogrid / woven GT 300 kN/m.

2.4 Global (rotational) stability

- Use **modified Bishop** with undrained strength (total stress) for short term; reinforcement force T acts at its level and **adds a resisting moment $T \cdot y$**.
- Unreinforced FS_u from search of critical circle (e.g. 0.9). Required extra tension:

$$T_{req} = (FS_{req} - FS_u) \cdot M_D / R$$

M_D = driving moment, R = lever arm (≈ radius if reinforcement is horizontal). The most critical circles often pass through the soft clay below the mattress, so also check deep circles and the reinforcement anchorage.

2.5 Other design checks

- **Reinforcement strain** at end of construction limited to ≈ 5–6 % (≤ 2 % residual for long-term); stiffness $J \geq T/\varepsilon$.
- **Embankment fill**: granular ($\varphi \geq 30°$), compacted ≥ 95 % MDD; side slopes protected against erosion.
- **Anchorage** length beyond the toe: wrap-around (return 1–2 m) or trench anchor.

2.6 Stage construction and strength gain

When the fill is raised in stages with waiting periods, pore pressure dissipates and the clay gains strength, which allows a higher final embankment:

$$\Delta c_u = (c_u / \sigma'_v) \cdot U \cdot \Delta\sigma'_v$$

Example 5

$c_u/\sigma'_v = 0.25$, $\Delta\sigma = 60$ kPa, $U = 70$ % after the first stage → $\Delta c_u = 0.25 \times 0.70 \times 60 = 10.5$ kPa

New $c_u = 15 + 10.5 = 25.5$ kPa → $H_c = 5.14 \times 25.5 / 19 = 6.9$ m (up from 4.1 m).

2.7 Reinforced steep side slopes

- Layers of geogrid / geotextile (spacing 0.3–0.6 m) with wrap-around or facing give slopes up to ≈ 70°, saving land width.
- Design like a wall: $T = K\sigma_v S_v$; check rupture, pull-out beyond the critical slip surface (log-spiral / circular), and local stability between layers.
- Secondary short layers near the face improve compaction and erosion resistance; use drainage and vegetation / erosion mats on the face.

3. Foundation Mattress below the Embankment

3.1 Purpose and function

- **Reinforcement** – takes tension, prevents lateral spreading and increases slope stability.
- **Load distribution** – spreads embankment load (about 2V:1H) so stress on soft soil is lower and more uniform.
- **Reduces differential settlement** and bridges local weak spots / voids.
- **Bearing capacity improvement** – tensioned-membrane effect and confinement.
- **Separation** – prevents mixing of fill and clay; **working platform** for equipment on very soft soil.
- **Drainage** – granular layer (with PVDs) provides a horizontal drainage path so consolidation is faster.

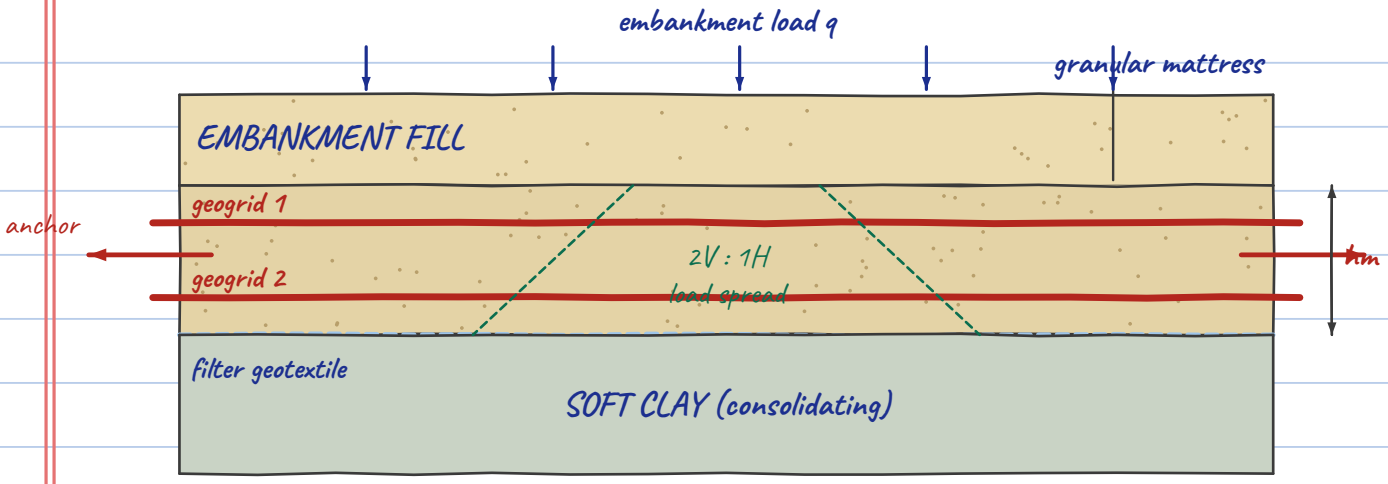


Fig 4.2 Reinforced granular foundation mattress

3.2 Components

1. **Geosynthetic reinforcement** – high-strength woven geotextile (PET / PP / PVA) or geogrid, 1–3 layers; strength 50–500 kN/m, low creep, strain < 5 %.
2. **Granular layer** – sand / gravel 0.3–1.0 m thick, $\varphi \geq 35^\circ$, compacted; also acts as drainage blanket.
3. **Separation / filter geotextile** (nonwoven) between soft clay and granular layer.
4. **Prefabricated vertical drains** (optional) connected to the blanket.
5. **Edge details** – wrap-around or anchor trench; toe berms; drainage outlets.

Variants: **geocell mattress** (3-D cells filled with aggregate, stiff slab-like), **fascine / brushwood** mattress (traditional), **piled mattress** over columns.

4. Design of Reinforced Mattress

Steps: (1) find T required for stability and spreading, (2) choose reinforcement and layers, (3) fix mattress thickness h_m, (4) check settlement, (5) bearing capacity, (6) long-term performance.

4.1 Mattress thickness and load spread

$$\Delta\sigma \text{ at clay top} = q\,B / (B + h_m) \quad \text{(2V:1H spread, strip load B)}$$

Typical h_m = 0.5–1.0 m; thicker layer → better load spreading but more weight; choose so that stress on soft clay stays below ≈ 5 c_u/FS.

4.2 Settlement control

Primary consolidation settlement of the soft clay (thickness H_0):

$$\rho_c = [C_c H_0 / (1 + e_0)] \cdot \log[(\sigma'_0 + \Delta\sigma') / \sigma'_0]$$

$$\rho_s = [C_\alpha H_0 / (1 + e_p)] \cdot \log(t_2/t_1) \text{ (secondary)}$$

Example 3

C_c = 0.45, e_0 = 1.2, H_0 = 6 m, σ'_0 = 40 kPa (mid-depth), $\Delta\sigma'$ = 60 kPa

ρ_c = (0.45 × 6 / 2.2) × log(100/40) = 1.227 × 0.398 = **0.49 m**

Time: $t = T_v H_{dr}^2 / c_v$; for U = 90 % (T_v = 0.848), c_v = 1.5 m^2/yr: single drainage (H_{dr} = 6) → 20 yr; double (3 m) → **5.1 yr**; with PVDs → months.

- Reinforcement **does not stop consolidation settlement**; it makes it **uniform**. Control by **PVDs, preloading / surcharge, stage construction and lightweight fill**.
- Allowed settlement: ≈ 100–300 mm total residual after pavement for highways; differential < 1:200 – 1:100.

4.3 Bearing capacity

$$q_{ult} = c_u N_c + \Delta q_R, \; N_c = 5.14, \; \Delta q_R \approx 2T \sin\theta / B$$

Δq_R = increase due to tensioned membrane / confinement (T = mobilised tension, θ = deflection angle at the edge of the loaded area, B = width). Use FS ≥ 1.3 (embankment, temporary) to 2.5 (permanent structures). Reinforcement mainly improves **stability and uniformity** of bearing rather than large strength increase.

Tensioned membrane over a soft spot / void of width B_s and sag d:

$$T = q_m B_s^2 / (8d), \; \varepsilon \approx (8/3)(d/B_s)^2$$

Example 4

q_m = 10 kPa, B_s = 2 m, d = 0.25 m → T = 10 × 4 / (8 × 0.25) = **20 kN/m**; ε = 2.67 × $(0.125)^2$ = **4.2 %** (< 5 % OK).

4.4 Long-term performance

- **Creep:** $T_{al} = T_{ult} / (RF_{CR} \cdot RF_{ID} \cdot RF_{D})$; PET preferred for sustained load; limit long-term strain to ≈ 2–3 %.
- **Durability:** pH, temperature, chemicals; hydrolysis (PET in alkaline soils), UV during construction.
- **Consolidation and secondary compression** – raise embankment (over-build) to compensate; add maintenance allowance.
- **Drainage clogging** – filter geotextile compatibility; blanket outlets kept open.
- **Strength gain** of clay with consolidation improves FS with time (stage construction exploits this).

4.5 Geocell and piled alternatives

- **Geocell mattress:** honeycomb cells (height 150–300 mm) filled with granular soil act like a stiff **slab**; confinement increases friction angle and spreads load widely – excellent over very soft clay (c_u < 10 kPa); thickness of mattress ≈ 0.3–0.6 m.
- **Piled embankment:** pile caps / stone columns with 1–3 layers of geogrid forming a **load-transfer platform**; arching in fill transfers load to piles; used where settlement must be almost nil (bridge approaches).
- Choice: basal geotextile + PVD (cheap, time available) → geocell (very soft, low fill) → piles (zero settlement, little time).

Quick recap

Design summary for a mattress: (1) T_{req} from spreading & circle → (2) choose $T_{al} \geq T_{req}$ (after reduction factors) → (3) h_m = 0.5–1 m granular → (4) check $q_{ult}/q \geq 1.3$ → (5) predict ρ and time (PVDs) → (6) specify monitoring and limits.

5. Field Implementation

1. **Site preparation:** clear vegetation, cut trees at ground level, **leave the root mat**; fill holes; avoid disturbing very soft soil (use light equipment / mats).

2. Place **nonwoven separator** and **sand blanket**; install **PVDs** and connect to the blanket (if used).
3. **Unroll geosynthetic** perpendicular to the embankment axis (main strength direction across the embankment), pull taut, remove wrinkles; **overlap 0.3–1 m** or sew joints (CBR < 1 → sew / 1 m).
4. **Anchor** with pins / sandbags; place **fill from the edges toward the centre** (end-tipping over a stable sub-layer) so a mud-wave is not formed; first lift ≥ **0.3–0.5 m**, use low-ground-pressure dozers; **no direct traffic on bare geosynthetic**.
5. **Compact** only after first lift thickness is placed; later lifts 200–300 mm to ≥ 95 % MDD; wrap-around at slopes; **stage the rate of filling** (e.g. 0.3–0.5 m/week) guided by monitoring.
6. Complete drainage, erosion protection of slopes and settle-out period before pavement.

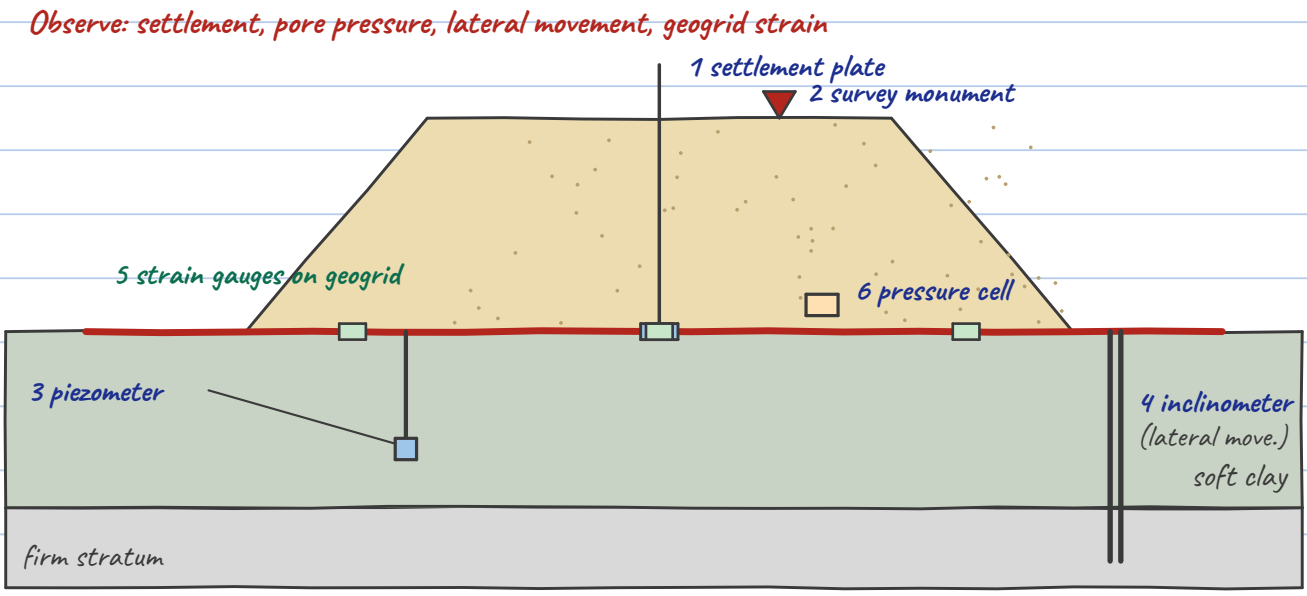


Fig 4.3 Instrumentation of a reinforced embankment

6. Monitoring Techniques

The **observational method** is used on soft soils: measure, compare with design predictions, and adjust the rate of filling.

Instrument	Measures	Purpose
Settlement plate / gauge	Vertical settlement at base	Degree of consolidation, Asaoka
Survey monuments (surface)	Heave, settlement at crest / toe	Differential movement
Piezometer (vibrating wire)	Excess pore pressure Δu	Rate of fill, strength gain
Inclinometer	Lateral displacement at toe	Stability, spreading
Strain gauge / tell-tale	Strain in geosynthetic	Tension T, creep, safety
Earth pressure cell	Stress at base	Load distribution
Extensometer	Differential vertical strain	Layer compression

Control criteria and prediction

- **Pore pressure ratio** $r_u = \Delta u / \Delta\sigma$ kept below ≈ 0.6–0.7 during filling; stop filling if it rises quickly.
- **Lateral-to-vertical movement** (Tavenas, Matsuo–Kawamura plots): a sudden increase of horizontal movement shows imminent failure; typical alert: $d\delta_h/dt >$ 2–3 mm/day.
- **Settlement prediction:** Asaoka method (plot ρ_i vs ρ_{i-1} at equal Δt; intersection with 45° line gives final settlement), hyperbolic method.
- **Geosynthetic strain** should remain < 5 %; creep strain should flatten with time.
- Frequency: daily during filling, weekly after, then monthly until settlement rate < 2–5 mm/month.

Advantages and limitations of reinforced embankments

- Allow construction on very soft soil (c_u < 15 kPa) without removal / replacement; **higher and steeper** embankments; faster construction and less fill.
- More uniform settlement, reduced cracking of pavement, less land take, lower cost than piling or soil replacement.

- Limitations: does not remove consolidation settlement; needs good quality control and monitoring; reinforcement cannot help if the failure surface is deep and passes below it.

Construction problems and remedies

Problem	Cause	Remedy
Mud-wave ahead of fill	Dumping on weak soil	Place from edges, thicker first lift
Wrinkles / slack	Poor tensioning	Pull taut, pin, sew joints
Geosynthetic damage	Direct traffic, sharp stones	Cover 0.3 m, select fill
High pore pressure	Fast filling	Stage filling, PVDs, wait
Edge failure / spreading	Weak anchorage	Wrap-around, longer anchorage

Case idea: embankment 6–8 m high on 8 m soft clay with PVDs at 1.5 m, two layers of 200 kN/m PET geotextile and a 0.6 m sand blanket; filled in two stages over 6 months and monitored by settlement plates and piezometers – a typical highway-approach solution.

Formula sheet – Unit IV

Item	Expression
Critical height	$H_c = 5.14\, c_u / \gamma$
Spreading tension	$T \geq FS \cdot \tfrac{1}{2} K_a \gamma H^2$
Extra tension (circle)	$T = (FS_{req} - FS_u)\, M_D / R$
Consolidation settlement	$\rho = C_c H_0 / (1+e_0) \cdot \log(\sigma'_1/\sigma'_0)$
Time	$t = T_v H_{dr}^2 / c_v$
Membrane tension	$T = q_m B_s^2 / (8\, d)$

Important questions

1. Explain the modes of failure of embankments on soft soil; how does reinforcement help?

2. What is a foundation mattress? Explain its purpose, function and components.
3. Describe the design of a reinforced mattress for settlement control and bearing capacity.
4. Discuss the long-term performance of reinforced embankments.
5. Explain field implementation and monitoring techniques with a neat sketch.

HANDWRITTEN STUDY NOTES

GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

Subject Code : 23A01702a

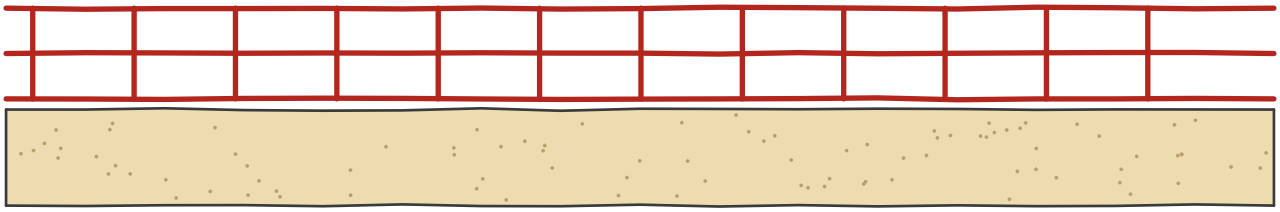
UNIT – V

Reinforced Soil Beds and Reinforced Pavements

Syllabus topics covered in these notes:

- Reinforced soil beds: introduction, factors affecting behaviour, analysis and design
- Reinforced pavements: benefits of reinforcement in flexible pavement layers
- Design of reinforced pavements by Giroud and Noiray approach
- Design of reinforced pavements by modified CBR method

soil + reinforcement = strong composite



1. Reinforced Soil Beds – Introduction

A **reinforced soil bed** (reinforced earth slab / reinforced foundation) is a layer of soil in which horizontal reinforcement (metal strips, geogrid, geotextile) is placed at several levels **below a shallow foundation** to increase its **bearing capacity** and reduce **settlement**. Pioneering work: **Binquet and Lee (1975)** with aluminium strips; later by Fragaszy and Lawton, Guido, Das, Khing, Omar and others using geogrids.

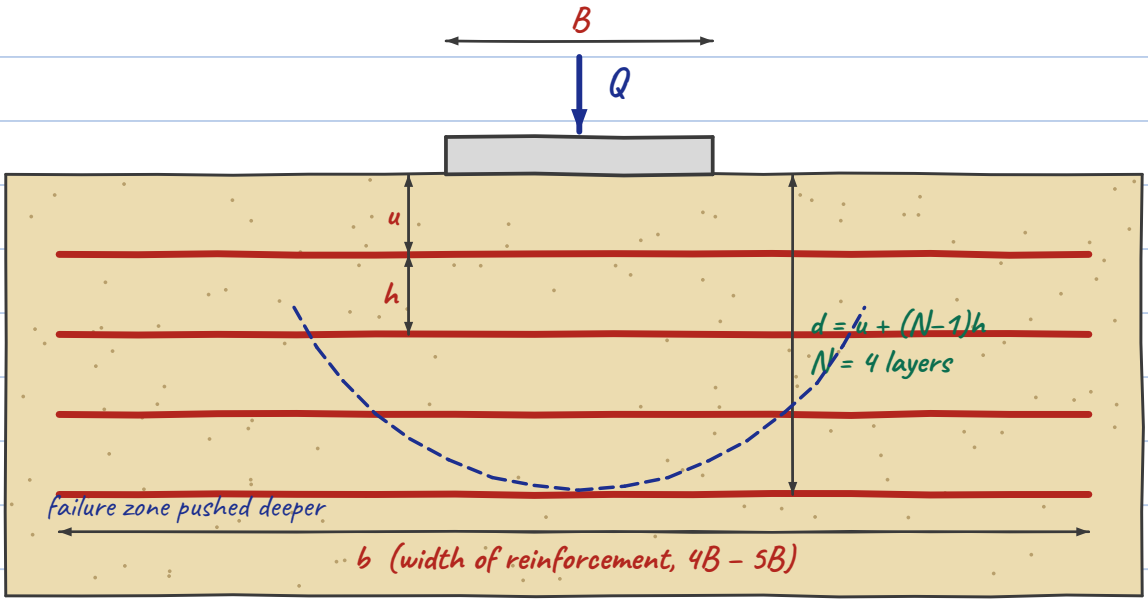


Fig 5.1 Footing on a reinforced soil bed

Geometric parameters

Symbol	Meaning	Typical optimum
B	width of footing	–
u / B	depth of first layer below footing	0.3 – 0.5 (≤ 2/3)
h / B	vertical spacing of layers	0.25 – 0.5
N	number of layers	3 – 5 (beyond: little gain)
b / B	width (length) of layers	4 – 5 (strip), 2.5–3 (square)
d / B	depth of reinforced zone d = u + (N–1)h	≈ 1.5 – 2

Mechanisms of improvement

1. **Deep-footing / rigid-boundary effect** – reinforced zone acts like a stiff slab, pushing the failure surface deeper into stronger or less stressed soil.
2. **Confinement / lateral restraint** – friction on the layers restrains lateral soil movement under the footing (enhanced confining pressure, pseudo-cohesion).
3. **Tensioned-membrane effect** – at large settlement the layers sag and their vertical component of tension carries part of the load.
4. **Load spreading** – wider and more uniform pressure on the lower soil.

2. Factors Affecting Behaviour of Reinforced Soil Beds

- **Depth of first layer (u/B)**: BCR is highest for $u/B \approx 0.3–0.5$; if $u/B > \approx 1$, the top layer is outside the stressed zone and gives almost no improvement (punching in the soil above).
- **Number of layers (N) and spacing (h/B)**: BCR increases with N, then levels off ($N \approx 4–5$); closer spacing is more effective but cost increases. Effective depth $d/B \approx 1.5–2$.
- **Length / width of reinforcement (b/B)**: must extend beyond the failure zone for pull-out resistance; BCR increases with b/B up to $\approx 4–5$ and then stays constant.
- **Type, strength and stiffness of reinforcement**: stiffer / stronger (steel strips, high modulus geogrid) gives higher BCR; too weak a layer **breaks**; geogrid aperture should match soil size (interlock).
- **Soil properties**: relative density and φ of the bed – benefit is higher for **loose sand** at small settlements; for very dense sand BCR is lower; cohesive soils need drainage layers.
- **Footing shape and size**: strip > square > circular in absolute gain; scale effect (small model footings overestimate BCR).
- **Surface roughness / interface friction** of reinforcement – higher friction, higher BCR.

- **Settlement level:** BCR is larger at larger settlement (s/B = 10–15 %) than at low settlement.

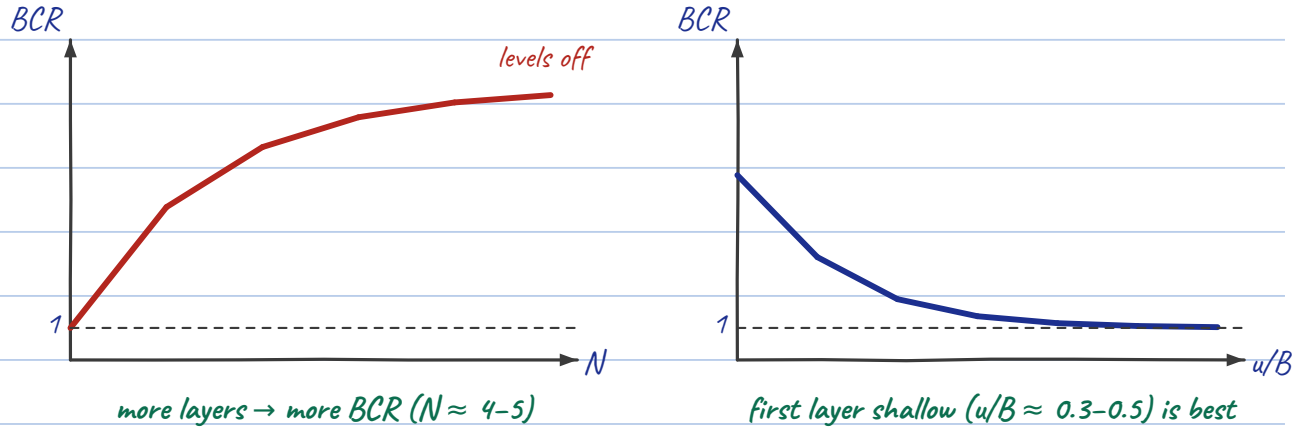


Fig 5.2 Typical trends of BCR (schematic)

Definitions

Bearing Capacity Ratio: $BCR = q_R / q_0$ (q_R = bearing pressure of reinforced bed, q_0 = unreinforced) at the same settlement or at failure.

Settlement Reduction Ratio $SRR = (\rho_0 - \rho_R)/\rho_0$ at the same load.

3. Analysis and Design of Reinforced Soil Beds

3.1 Failure modes (Binquet and Lee)

1. **Reinforcement breaks** (tension failure) – when layers are weak, many, and close to the footing (u/B small).
2. **Reinforcement pulls out** – when length is short or friction is low.
3. **Soil failure above the top layer** – when u/B is large (> 2/3): footing punches through the cover soil.
4. **General shear below the reinforced zone** – when d/B is too small.

3.2 Design procedure

1. Find q_0 (unreinforced) of the foundation soil: $q_0 = cN_c + qN_q + \frac{1}{2}\gamma B N_\gamma$, and the settlement it would give.
2. Choose the geometry: u/B = 0.3–0.5, h/B = 0.3–0.5, N = 3–5, b/B ≥ 4 (strip) → reinforced depth d.

3. Estimate **BCR** from design charts (Binquet and Lee) / test data: $q_R = BCR \cdot q_0$; $q_{all} = q_R / FS\ (\geq 3)$.
4. **Tension in layers:** compute from the load spread (T_{max} at the layer below the footing edge); check $T_{al} \geq T_{max}$ after reduction factors, with $FS \geq 1.5–2$ against breakage.
5. **Pull-out:** $L_e \geq FS \cdot T / (2 \mu^* \sigma_v)$ beyond the failure zone ($L_e \geq 1\ B$ typical); total width b = footing zone + $2 L_e$.
6. Check **settlement** (SRR) and the strength of the underlying soil at depth d (stress spread 2V:1H).

Example 1

Unreinforced strip footing: $q_0 = 300$ kPa. Reinforced bed $u/B = 0.4$, $h/B = 0.4$, $N = 4$, $b/B = 5 \rightarrow$ BCR (chart) ≈ 2.2

$q_R = 2.2 \times 300 = 660$ kPa; $q_{all} = 660 / 3 = 220$ kPa (instead of 100 kPa) → footing width can be reduced by ≈ 55 %.

3.3 Pseudo-cohesion approach (idea)

Treat the reinforced zone as soil with apparent cohesion $c_R = (T / 2S_v)\sqrt{K_p}$ (Unit I) and use a two-layer bearing capacity analysis; **use the tension mobilised at small strain (2–5 %)**, not T_{ult}. Numerical (FEM) and Binquet–Lee design charts are the usual practical tools.

Example 2

Sand $\varphi = 35°$: $K_p = 3.69$, $\sqrt{K_p} = 1.92$; geogrid mobilised $T = 10$ kN/m at $S_v = 0.4$ m →

$c_R = (10 / 0.8) \times 1.92 = 24$ kPa (equivalent apparent cohesion of the reinforced zone).

3.4 Settlement behaviour and design checks

- Load–settlement curves of reinforced beds are **stiffer**; the improvement appears mostly **after** the settlement exceeds $\approx$ 2–3 % of B, when the layers are mobilised.

- For design, limit settlement to the permissible value (e.g. 25–50 mm isolated footing) and find q from the reinforced curve at that settlement; apply FS only to the ultimate value.
- Reinforced layers should be **extended and anchored** beyond the footing edges; avoid tearing at the footing edge by using a thin sand cushion for geogrids on angular aggregate.
- In **cohesive soils** replace 1–2 m of weak soil with a compacted granular reinforced bed (sand/gravel mattress with 3–4 geogrid layers); provide drainage.

Example 3 (tension)

Tension check: footing load $q = 220$ kPa on $B = 2$ m; layer at depth $z = 1.0$ m; stress spread 2V:1H → $\Delta\sigma_v = qB/(B + z) = 220 \times 2/3 = 147$ kPa

Assume lateral thrust ≈ $K_a \Delta\sigma_v S_v = 0.27 \times 147 \times 0.8 = 31.8$ kN/m → choose geogrid with $T_{al} \geq 32$ kN/m (i.e. $T_{ult} \approx 32 \times 2.4 \approx 77$ kN/m for PET).

Applications of reinforced soil beds

- Shallow footings, rafts and **tank / silo foundations** on weak sand, silt or fill; embankment and abutment foundations.
- Footings on **expansive / collapsible** soil (with replacement and reinforcement); **machine foundations**; temporary working platforms.
- Strengthening of **strip footings near slopes** (reinforcement increases edge-distance capacity).

4. Reinforced Pavements

Flexible pavements on **weak subgrade** (CBR < 3–4 %) need thick granular layers. Placing a **geogrid or geotextile** at the subgrade–base interface (or within the base / asphalt) gives a more economical and longer-lasting pavement.

4.1 Benefits of reinforcement in flexible pavement layers

1. **Lateral restraint of base aggregate** – interlock with geogrid reduces spreading of aggregate, increasing base stiffness and modulus.

2. **Improved bearing capacity** – failure surface moves into the base; stress on subgrade is reduced (N_c rises from π to $\pi + 2$).
3. **Tensioned-membrane effect** – with rutting, the curved sheet carries load.
4. **Separation / filtration** – prevents subgrade fines pumping into aggregate and keeps the design thickness (geotextile).
5. **Reduced thickness** – base reduction **30–50 %** (aggregate thickness reduction, ATR) for same traffic.
6. **Longer life – Traffic Benefit Ratio TBR** = $N_{reinforced} / N_{unreinforced}$ (typically 1.5 – 10).
7. **Reduced rutting** and **fatigue / reflective cracking** (reinforcement in asphalt layers, geogrid / glass-fibre grid).
8. **Construction platform** on very soft soils (haul roads, working platforms); faster construction; lower life-cycle cost and carbon.

4.2 Types of geosynthetics used in pavements

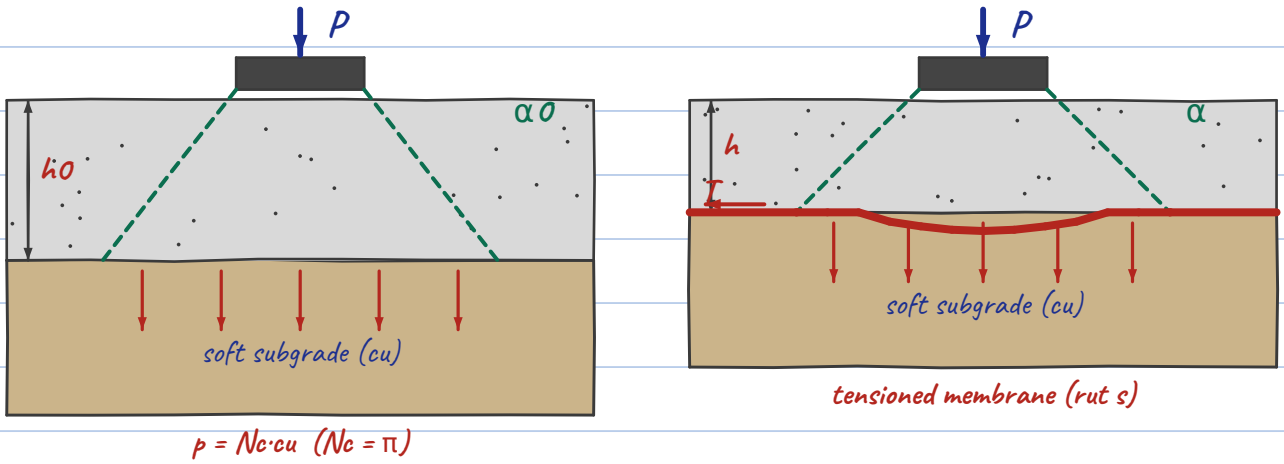
Location	Product	Main function
Subgrade – base interface	Woven GT, biaxial/triaxial geogrid	Separation, reinforcement
Within granular base	Geogrid, geocell	Lateral restraint, stiffness
Under / within asphalt	Glass-fibre, PET or steel grid; paving fabric	Anti-reflective cracking, rutting
Edge / side drain	Nonwoven GT, geocomposite	Filtration, drainage

- Guidelines: **IRC:SP:59** (geotextiles in road pavements), **IRC:37** (flexible pavement design), AASHTO R50 / FHWA manuals.
- Reinforcement is most effective when subgrade CBR is **1–5 %** and traffic is moderate-to-high; for CBR > 8 % benefits are small.
- Benefits are verified by **plate load tests, field trial sections and accelerated pavement testing** (rut depth vs passes).

Example 4

Example benefit calculation: unreinforced base 450 mm; reinforced base 300 mm → ATR = 33 %.

Unreinforced life 10 000 passes; reinforced life 40 000 passes at the same thickness → TBR = 4.



(a) Unreinforced – thicker base (b) Reinforced – thinner base

Fig 5.3 Load spread – unreinforced and reinforced base

Positions: (a) **subgrade / base interface** – most common; (b) **within the base** (upper third) for stiffening; (c) **within or beneath asphalt** (anti-cracking grid).

For subgrade CBR < 1–2 %, use a high-strength geotextile + geogrid combination.

5. Giroud and Noiray (1981) Approach

Design method for **geotextile-reinforced unpaved / haul roads** (also used for base courses of pavements). Based on **load spread through the aggregate** and the bearing capacity of the subgrade, with **tensioned-membrane** action for the reinforced case.

Assumptions

- Wheel load P on circular contact of radius r ($r = \sqrt{[P/(\pi\, p_c)]}$, p_c = tyre pressure) spreads through the aggregate at **angle** α_0 (unreinforced) and α (reinforced, $\geq \alpha_0$).

- Stress on subgrade $p = P / [\pi (r + h \tan\alpha)^2] = p_c / (1 + h \tan\alpha / r)^2$.
- Allowable subgrade stress: **unreinforced** $p = N_c c_u$, $N_c = \pi$ (3.14); **reinforced** $N_c = \pi + 2$ (5.14) because the geotextile prevents the lateral bulge, plus membrane term Δp.
- Undrained strength of subgrade $c_u \approx 30$ CBR (kPa).

Aggregate thickness

$$h = r [\sqrt{(p_c / (N_c c_u))} - 1] / \tan\alpha$$

- **Unreinforced:** h_0 with $N_c = 3.14$ and α_0 (depends on traffic passes N and allowed rut depth; $\alpha_0 \approx 38°$ for a few passes, smaller for heavy traffic – take from design chart).
- **Reinforced:** h with $N_c = 5.14$ (and α); if the rut is large, the geotextile curvature gives a pressure reduction $\Delta p = T / R_c$ ($R_c \approx b^2 / 8s$ radius of the deformed sheet, rut depth s, width b) and strain $\varepsilon \approx (8/3)(s/b)^2$; hence **required tension** $T = \Delta p \cdot R_c$.
- **Savings** $= h_0 - h$; compare with geotextile cost.

Example 5

Single tyre load $P = 40$ kN, $p_c = 480$ kPa → $r = \sqrt{[40/(\pi \times 480)]} = 0.163$ m.

Subgrade CBR = 1 → $c_u \approx 30$ kPa. Take $\alpha_0 = \alpha = 38°$ (tan = 0.781).

Unreinforced: $N_c c_u = 3.14 \times 30 = 94.2$ kPa; $\sqrt{(480/94.2)} = 2.257$ → $h_0 = 0.163 \times (2.257 - 1)/0.781 =$ **0.26 m**

Reinforced: $N_c c_u = 5.14 \times 30 = 154$ kPa; $\sqrt{(480/154)} = 1.764$ → $h = 0.163 \times 0.764/0.781 =$ **0.16 m**

Saving ≈ 0.10 m (≈ 39 %). (Illustration only – real design uses α from the chart for the required number of passes.)

Design steps (Giroud–Noiray)

1. Find wheel load P, tyre pressure p_c, number of passes N, allowed rut depth s.
2. Find subgrade c_u (CBR, vane, CPT).
3. Compute r; choose α_0, α from charts.
4. Compute h_0 ($N_c = \pi$) and h ($N_c = \pi + 2$).

5. Compute required geotextile **tension T** from rut geometry and strain; select product with $T_{al} \geq T$ and compatible modulus.
6. Check filtration, survivability, overlap / lap (0.3–1 m) and construction details.

Giroud–Han (2004) – modern improvement

Giroud and Han extended the approach to **geogrid-reinforced unpaved roads** using a stiffness-dependent bearing factor N_c and a load-spreading angle that depends on **aperture stability modulus**], base thickness and rut depth; the design is based on required base thickness for given traffic N passes and rut depth.

6. Modified CBR Method

The **CBR method** (U.S. Corps of Engineers) gives the pavement / base thickness from the wheel load, tyre pressure and subgrade CBR. In the **modified CBR method** for reinforced pavements, the geosynthetic is accounted for by using a **modified (improved) CBR** of the subgrade, $CBR_m = IF \times CBR_{sg}$, where IF = improvement factor from laboratory CBR tests on reinforced specimens (geosynthetic placed in the mould) or from plate-load / DCP tests; thickness is then read from the standard design curves (IRC:37 / IRC:SP:59) or the formula.

$$t = \sqrt{\{P\,[\,1/(8.1\,CBR) - 1/(\pi\,p)\,]\}}$$ (P in lb, p in psi, t in inch)

$$\text{SI: } t = \sqrt{\{P\,[\,1/(55.9\,CBR) - 1/(\pi\,p)\,]\}}$$ (P in kN, p in kPa, t in m)

Procedure

1. Determine soaked **CBR of subgrade** (unreinforced) – CBR_{sg}.
2. Determine CBR_m with the selected geosynthetic (laboratory reinforced CBR or field test): $IF = CBR_m / CBR_{sg}$ (typically ≈ 1.3–2 for geogrids on weak soil – obtain from tests).
3. Design traffic: wheel load P, tyre pressure p (or MSA for IRC:37).
4. Compute thickness t_0 with CBR_{sg} and t_R with CBR_m.
5. Thickness reduction = $t_0 - t_R$ (percentage savings); check minimum layer thicknesses and drainage.

Example 6

$P = 40$ kN, $p = 480$ kPa, $\pi p = 1508$; $CBR_{sg} = 2$ %, $CBR_m = 3.5$ %

Unreinforced: $1/(55.9 \times 2) = 0.008945$; – $1/1508 = 0.000663 \rightarrow 0.008282 \times 40 = 0.331 \rightarrow t_0 = 0.576$ m

Reinforced: $1/(55.9 \times 3.5) = 0.005111 - 0.000663 = 0.004448 \times 40 = 0.178 \rightarrow t_R = 0.422$ m

Saving = 0.154 m ≈ 27 % of the thickness; IF = 1.75.

Giroud–Noiray vs Modified CBR

Point	Giroud–Noiray	Modified CBR
Basis	Load spread + bearing capacity	Empirical CBR equation / charts
Subgrade property	cu (≈ 30 CBR)	CBR (modified CBRm)
Reinforcement effect	Nc π → π+2, membrane	Improvement factor IF
Product data	Tension, modulus	Test-derived IF
Use	Unpaved / haul roads	Paved flexible pavements

Construction notes

- Trim and compact the subgrade; remove sharp objects; roll out the geogrid / geotextile **flat and taut**; lap **0.3–1 m** (CBR-dependent) or join mechanically.
- Spread aggregate by **end-dumping from the edge** onto the sheet; first lift ≥ 150–300 mm; no turning of tracked vehicles on the bare sheet; compact to ≥ 98 % MDD / specification.
- Provide edge anchorage and drainage; in asphalt reinforcement, use tack coat and compatible grid temperature rating.

Formula sheet – Unit V

Item	Expression
Bearing capacity ratio	$BCR = qR / q0$
Layer depth	$d = u + (N - 1)\,h$
Subgrade stress	$p = P / \pi\,(r + h \tan\alpha)^2$
Aggregate thickness	$h = r(\sqrt{(pc/Nc\ cu)} - 1)/\tan\alpha$
Nc	π (unreinforced); $\pi + 2$ (reinforced)
CBR thickness (SI)	$t = \sqrt{\{P[1/(55.9\ CBR) - 1/(\pi\ p)]\}}$

Important questions

1. What is a reinforced soil bed? Explain mechanisms of improvement of bearing capacity.
2. Discuss the factors affecting the behaviour of reinforced soil beds (u/B, N, h/B, b/B).
3. List the benefits of placing reinforcement in flexible pavement layers.
4. Explain Giroud and Noiray design approach for reinforced pavements with an example.
5. Explain the modified CBR method of design of reinforced pavements.